Econometrics and Statistics 8 (2018) 204–230

Contents lists available at ScienceDirect

Econometrics and Statistics

journal homepage: www.elsevier.com/locate/ecosta

A new particle filtering approach to estimate stochastic volatility models with Markov-switching

Frédéric Karamé

Le Mans University, GAINS-TEPP (FR CNRS 3435), Institut du Risque et de l'Assurance du Mans and DynareTeam (Cepremap)

ARTICLE INFO

Article history:
Received 18 January 2017
Revised 23 May 2018
Accepted 24 May 2018
Available online 22 June 2018

Keywords:
Stochastic volatility
Bayesian inference
Markov switching
Particle filtering
Smooth resampling
Kim filter
nonlinear Kalman filter

ABSTRACT

A simple method is proposed to estimate stochastic volatility models with Markov-switching. It relies on a nested structure of filters (a Hamilton filter and several particle filters) to approximate unobserved regimes and state variables, respectively. Smooth resampling is used to keep the computational complexity constant over time and to implement a standard likelihood-based inference on parameters. A bootstrap and an adapted version of the filter are described and their performance are assessed using simulation experiments. The volatility of US and French markets is characterized over the last decade using a three-regime stochastic volatility model extended to include a leverage effect.

© 2018 EcoSta Econometrics and Statistics. Published by Elsevier B.V. All rights reserved.

1. Introduction

Nonlinear models can address various issues in finance, particularly in accounting for volatility and risk. Since the seminal works of Engle (1982) and Bollerslev (1986), many studies extended ARCH/GARCH models. Indeed, squared returns capture volatility in these representations, which renders their estimations relatively feasible even in complicated versions. Stochastic volatility models (SV hereafter, Taylor, 1986) provide an alternative approach. These representations are flexible and more general, as a latent variable captures volatility, rather than being only conditioned by prices (see Wiggins, 1987; Hull and White, 1987; Heston, 1993). However, the presence of this latent variable complicates the estimation, as the wide variety of methods proposed in the literature show (estimation of approximated models (Harvey et al., 1994; Langrock et al., 2012), indirect inference (Smith, 1993; Gourieroux et al., 1993; Gallant and Tauchen, 1996; Gallant et al., 1997; Sandmann and Koopman, 1998), or Monte-Carlo Markov Chain methods (*aka* MCMC, (Jacquier et al., 1994; Chib et al., 2002), or (Johannes and Polson, 2002)).

A natural way to estimate nonlinear/non-Gaussian representations with unobserved variables is to use Sequential Monte Carlo methods, *i.e.*, particle filtering (Doucet et al., 2001; Arulampalam et al., 2002; Chen, 2003; Cappe et al., 2007; Creal, 2009), or (Doucet and Johansen, 2009). In this case, the likelihood is not available in a closed form and is approximated. However, the resampling step, which is necessary to avoid degeneracy in a large sample, makes inference through maximum likelihood using standard techniques difficult (Pitt, 2002; Kantas et al., 2015). Indeed, the likelihood estimator depends on both the resampled particles and unknown parameters, which induces discontinuity in the estimation criterion. Malik and Pitt (2011) propose a smooth resampling approach that produces an estimator for the likelihood that is continuous with

E-mail address: frederic.karame@univ-lemans.fr

https://doi.org/10.1016/j.ecosta.2018.05.004
2452-3062/© 2018 EcoSta Econometrics and Statistics. Published by Elsevier B.V. All rights reserved.

respect to the unknown parameters (see also Lee, 2008, Liesenfeld and Richard, 2003, and Liesenfeld and Richard, 2006 for alternative approaches). Direct parameter inference on this type of model become possible through maximum likelihood.

So et al. (1998) propose enhancing the SV model and introducing Markov switching for some parameters (MS-SV hereafter) to capture volatility clusters observed in financial data, *i.e.*, the regimes of high and low volatilities, and to improve estimates of the volatility persistence at the same time (Lamoureux and Lastrapes, 1990). Technically, it amounts to conditioning some parameters of the model to an extra discrete unobserved variable that captures the regimes. Implementing the standard particle filter with smooth resampling is probably the most automatic and straightforward approach to estimate MS-SV models (see also So et al., 1998, Smith, 2002, Hwang et al., 2004, or Langrock et al., 2012) who estimate an approximated version of the model through standard maximum likelihood and (Chen and Liu, 2000), (Fearnhead and Clifford, 2003), or (Carvalho and Lopes, 2007) who use particle-filtering based methods). However, smooth resampling in a multivariate context and in the presence of a discrete latent variable is delicate and will probably generate discontinuities in the likelihood. It could require a large number of particles (and consequently, large computation times) to obtain a smooth likelihood surface.

Thus, we develop a new tool to estimate nonlinear/non-Gaussian models with Markov switching by maximum likelihood. We implement this tool for the MS-SV model, originally proposed by So et al. (1998) and extended to incorporate a leverage effect (MS-SVL hereafter) because it provides a natural illustration of this issue. We show that it is fast, displays very good performance for both latent variables and parameter estimation, does not require model-dependent specific calculations, and has the advantage of combining only well-known existing tools. Additionally, this approach could be extended to nonlinear models, including multiple Markov chains and/or a multivariate set of continuous state variables with an efficient version of multivariate smooth resampling (see Gerber and Chopin, 2015 who sort a multivariate set of state variables with Hilbert curve mapping). For instance, introducing a Markov-switching mechanism in a DSGE model can explicitly capture a change in economic policy or in agents' behavior (the central banker in a Taylor rule, active/passive monetary/fiscal policies, etc.) or a structural break in the economy (such as entering the Zero Lower Bound on nominal interest rates) (Bianchi, 2013; Liu et al., 2011). This literature rapidly evolved from linearly solved MS-DSGE models (Liu and Mumtaz, 2011; Alstadheim et al., 2013) to nonlinearly solved MS-DSGE models (Dueker et al., 2002; Foerster et al., 2016; Binning and Maih, 2015; 2016; Albertini and Lan, 2016), opening the way for promising future applications. Finally, this method can also be viewed as a special case of the recent and promising approach of nested filters (Chopin et al., 2013; Crisan and Miguez, 2016).

This study has two main contributions. First, we do not approximate the regime variables by a swarm of particles, as in Chen and Liu (2000), Fearnhead and Clifford (2003), or Carvalho and Lopes (2007). We rather exploit the Bayesian inference on regimes analytically to handle regime probabilities directly. Hence, we circumvent the problem of smooth resampling a discrete variable. We examine every possible transition at each time and conserve only the current information on regimes, which confers a two-layer structure to our filter: a typical Hamilton-like filter (the outer part that will handle the regime probabilities) and nesting particle filters for each possible transition between previous and current regimes (the inner part). This is the same structure as in Kim (1994), except that we use particle filters for each regime transition to approximate the current log-volatility and estimate the conditional likelihood because the conditional state-space representation is nonlinear/non-Gaussian. Second, we use univariate smooth resampling to achieve three aims simultaneously: *(i)* update the particle swarm; *(ii)* feed the next time iteration of the filter with only one constant-sized swarm of particles by regime, which prevents the algorithm from exploding after several iterations due to the multiplication of particle swarms by the number of regimes at each time iteration; and *(iii)* implement a likelihood-based inference on parameters.

The paper is organized as follows. Section 2 discusses the MS-SVL model and the difficulties in using the standard smooth particle filter. Section 3 discusses our approach. We focus on the choice of the proposal distribution for the log-volatility. Beyond the usual 'blind' choice based on the transition density, we use the Gaussian *posterior* density from a nonlinear Kalman filter, as in Kotecha and Djuric (2003), (Winschel and Kratzig, 2010), or (Murray et al., 2013). Section 4 compares their performance with (i) the standard smooth particle filter, (ii) our implementation of the maximum-likelihood based HMM approach proposed by Langrock et al. (2012), and (iii) the online filter proposed by Carvalho and Lopes (2007). We retain two dimensions for the analysis. First, we estimate a simulated dataset to compare and discuss the different methods. Second, we conduct a Monte-Carlo experiment to assess the properties of the parameter estimators, the fit of the log-volatility, and the regime occurrences. For the adapted version, we show that our approach is fast and accurate when implemented for the MS-SVL model. It provides a very good fit for both volatility and regime occurrences because our probabilities are smoothed; that is, they incorporate all of the sample information. The performance in parameters estimation is also good, despite the low number of particles. We show that in its adapted version, our approach globally outperforms the benchmark methods and significantly improves the leverage effect estimation. Finally, Section 5 discusses the estimation results for real data. We study the S&P500, the NASDAQ, and the CAC40 indices over a recent period (January 2006 to June 2015) using a three-regime MS-SVL model to capture low, high, and very high levels of volatility. Section 6 concludes the paper.

2. Framework

This section presents the MS-SVL model and the main elements of particle filtering. Further, we discuss the implementation of the smooth particle filter for the MS-SVL model.

2.1. Markov-switching stochastic volatility model with leverage

The MS-SVL model can be defined as follows:

$$\begin{aligned} r_t &= e^{\frac{h_t}{2}}\ \varepsilon_t, \\ h_{t+1} &= \mu_{s_{t+1}}(1-\phi) + \phi h_t + \sigma_\eta \eta_t, \\ \begin{pmatrix} \varepsilon_t \\ \eta_t \end{pmatrix} &\sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right), \end{aligned} \tag{1}$$

where $\{r_t\}_{t=1}^T$ is the observed returns and $\{h_t\}_{t=1}^T$ the unobserved log-volatility. $s_t \in \{1, \ldots, S\}$ is an unobserved one-order Markov chain of S regimes, with fixed transition probabilities $p(s_t = j | s_{t-1} = i) = p_{ij}$ and $\sum_{j=1}^S p_{ij} = 1, \quad \forall (i,j) \in \{1, \ldots, S\}^2$. ε_t and η_t are Gaussian $\mathcal{N}(0,1)$. $\Theta = \{\mu_1, \ldots, \mu_S, \phi, \sigma_\eta, \rho, p_{ij}\}$ is the global set of unknown parameters with $(i,j) \in \{1, \ldots, S\} \times \{1, \ldots, S-1\}$.

We can see this model as an extension of the canonical SV model to capture stylized facts that characterize financial data. First, we measure the leverage effect; that is, the negative correlation between the shocks on volatility and returns (or prices), by parameter ρ. Recent studies estimate this representation for several stock market indices such that the SV model with leverage seems to be the new canonical model. They find highly significant leverage effects (see, for instance, Raggi and Bordignon, 2006; Omori et al., 2007; Bollerslev et al., 2009; Nakajima and Omori, 2009; Malik and Pitt, 2011, or Abanto-Valle et al., 2011). Second, others introduce Markov-switching regimes to capture the clusters of volatility. Lamoureux and Lastrapes (1990) argue that the persistence parameter ϕ can be overestimated if changes in regimes are ignored in the volatility process. A popular technique to capture these regimes is to allow a discrete shift in the drift of the volatility process using the regime-dependent intercept μ_{s_t} through a Markovian process $s_t \in \{1, \ldots, S\}$ (see, e.g., Cai, 1994 or Hamilton and Susmel, 1994 for an ARCH-type model). The transition probabilities p_{ij} capture the persistence of the volatility regimes more efficiently than the single parameter ϕ does (see So et al., 1998).

2.2. Smooth particle filter

Suppose a non-linear and non-Gaussian state-space form

$$\begin{aligned} x_t &= f(x_{t-1}, \xi_t; \Theta), \\ y_t &= g(x_t, \varepsilon_t; \Theta), \end{aligned} \tag{2}$$

where y_t is the set of observed (measurement) variables; x_t the set of unobserved state variables; ε_t and ξ_t are the measurement and state innovations, respectively, *iid* $\mathcal{N}(0,1)$. Θ is still the global set of unknown parameters.

The Bayesian approach teaches us to infer $p(x_t | y_{1:t}; \Theta)$, the density of the state variables at date t conditional on $y_{1:t}$, the current and past information on the observables, given Θ. A simple and powerful solution to estimate this type of model is to use particle filtering. The idea is to choose an easy-to-sample proposal distribution $q(x_t | y_{1:t}; \Theta)$, informative on the target distribution $p(x_t | y_{1:t}; \Theta)$, such as

$$q(x_t | y_{1:t}; \Theta) = q(x_t | x_{t-1}, y_t; \Theta) \times q(x_{t-1} | y_{1:t-1}; \Theta). \tag{3}$$

The filter can be implemented sequentially by randomly drawing the current particles $\{\tilde{x}_t^{(i)}\}_{i=1:M}$ in $q(x_t | x_{t-1}^{(i)}, y_t; \Theta)$ at each date, given the previous state distribution $p(x_{t-1} | y_{1:t-1}; \Theta)$ is approximated with the particle swarm $\{x_{t-1}^{(i)}, w_{t-1}^{(i)}\}_{i=1:M}$. The particle (unormalized) weights are

$$\hat{w}_t^{(i)} \propto w_{t-1}^{(i)} \frac{p(y_t | \tilde{x}_t^{(i)}; \Theta)\, p(\tilde{x}_t^{(i)} | x_{t-1}^{(i)}; \Theta)}{q(\tilde{x}_t^{(i)} | x_{t-1}^{(i)}, y_t; \Theta)}, \tag{4}$$

which depend on their parent weight and the incremental weight. A first approximation is then $\left\{ \tilde{x}_t^{(i)}, \tilde{w}_t^{(i)} = \frac{\hat{w}_t^{(i)}}{\sum_{i=1}^M \hat{w}_t^{(i)}} \right\}_{i=1:M}$, where $\tilde{w}_t^{(i)}$ is the normalized weight. Finally, we can approximate the evidence by

$$p(y_t | y_{1:t-1}; \Theta) \approx \sum_{i=1}^M \hat{w}_t^{(i)}. \tag{5}$$

Given that weights degenerate as t increases to the point where all-but-one particle have negligible weights (see, e.g., Doucet et al., 2001). Essentially, this pattern is valid for large samples, such as in finance. Thus, early studies initially proposed systematic resampling (Gordon et al., 1993). It consists of randomly drawing with replacement particles from their empirical distribution approximated by $\{\tilde{x}_t^{(i)}, \tilde{w}_t^{(i)}\}_{i=1:M}$. It amounts to discarding particles with low weights and replicating particles with high weights to focus on interesting areas of the distribution using a constant number of particles. Consequently, all new particles have the same weight afterwards $\{x_t^{(i)}, w_t^{(i)} = \frac{1}{M}\}_{i=1:M}$.

Beyond the impoverishment of the particle swarm and the computational burden, the most important limit of resampling is that it complicates inference through maximum likelihood using standard techniques. Even when the seed for random

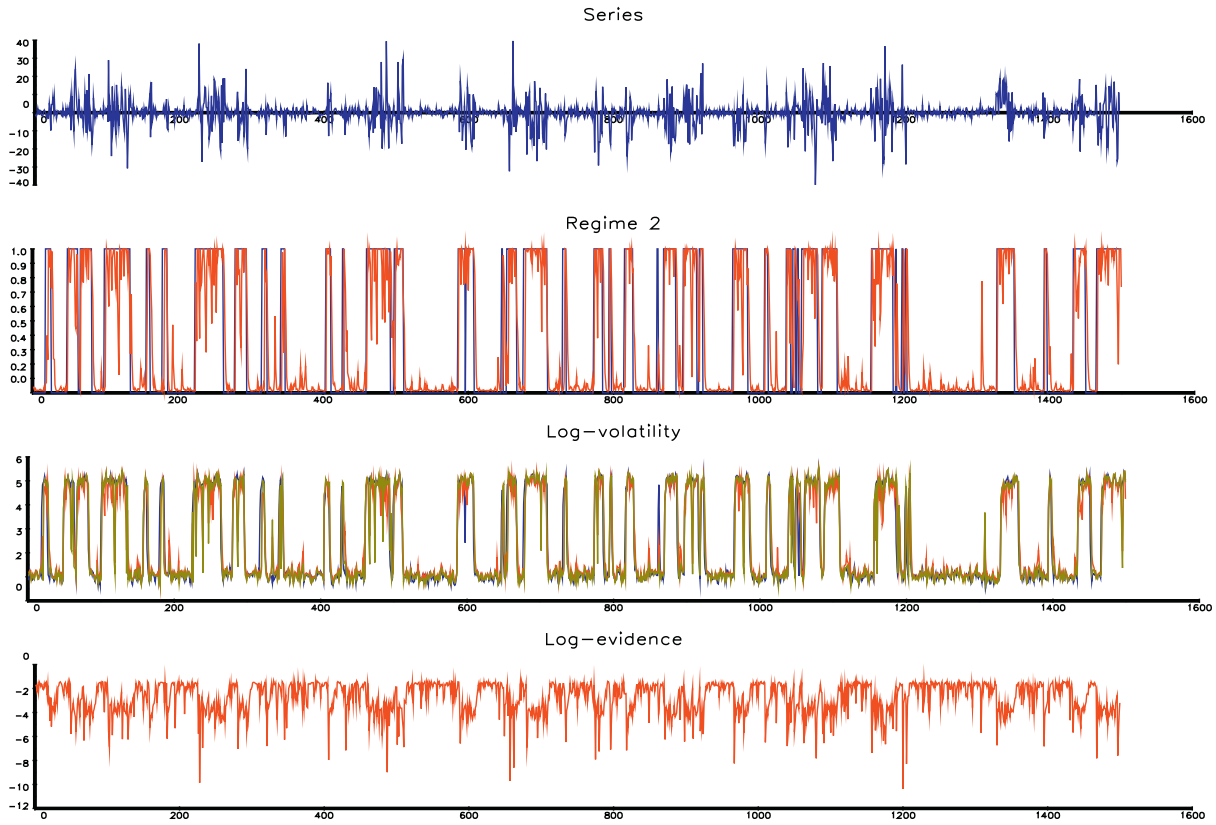


Fig. 1. Unobserved components provided by the smooth SIR (case 1): (MS-SVL model, $M = 50{,}000$). (a) Simulated returns (b) Regime 2 (true occurrence (in blue) vs. filtered probabilities (in red)) (c) Log-volatility (true (blue) vs. mean (red) and median (green) estimates) (d) Estimated log-evidence (in red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

draws is fixed across the simulations, the traditional likelihood estimator depends on both the resampled particles and unknown parameters. A small change in the parameter value will cause a small change in the importance weights, which will potentially generate a different set of resampled particles. This produces a discontinuity in the likelihood criterion and explains why applied studies depart from the usual maximum likelihood approach. Malik and Pitt (2011) propose using smooth resampling to produce a likelihood estimator that is continuous in the parameters (see Appendix A for details). Parameter inference is then possible through simulated maximum likelihood using particle filtering (see Appendix B-1 for details).

2.3. Implementation for the MS-SVL model

Suppose we now cast the MS-SVL model (1) into the non-linear and non-Gaussian state-space representation (2), where y_t is equal to r_t, and $x_t = \{h_t, s_t\}$ is the set of unobserved state variables. The state equation can be reformulated as:

$$h_{t+1} = \mu_{s_{t+1}}(1-\phi) + \phi h_t + \sigma_\eta \rho y_t e^{-\frac{h_t}{2}} + \sigma_\eta \sqrt{1-\rho^2}\ \xi_t. \tag{6}$$

The conditional likelihood is then

$$p(y_t|h_t;\Theta) = \frac{1}{\sqrt{2\pi e^{h_t}}} e^{-\frac{1}{2}\frac{y_t^2}{e^{h_t}}}.$$

Implementing smooth resampling raises two difficulties. First, smooth resampling two state variables is relatively difficult. In the multivariate case, (Malik and Pitt, 2011) propose the following approach. First, the uniform line is divided into P fixed partitions (or quantiles). The particle swarm is sorted along one state variable and each particle is attached to one partition along its cumulative density function. The weights are then normalized within each partition. Finally, univariate smooth resampling is implemented for each variable in each partition to produce a swarm of M/P particles. This approach is relatively burdensome, but attempts to preserve the relationship between the state variables for P large. To circumvent this issue, we assume "perfect correlation" between the log-volatility and the regime occurrence, meaning that particles with low log-volatility values are more likely associated with regime 1, while particles with high log-volatility values are more likely associated with regime 2. Consequently, we sort the particles on both state variables only along the log-volatility, and

implement smooth resampling for each state variable as in the univariate case. This approach can be viewed as natural and reasonable for this model and considerably reduces the computation time.

The second difficulty is more crucial. The presence of the regime variable s_t raises a major problem due to its discrete nature because the smooth resampling interpolates sorted particles. How do we apply this method to the regime (discrete) variable when two particles do not belong to the same regime? Even with a limited occurrence, this issue is important since some particles of the discrete variable can still have different values for common parts of the log-volatility support, where conditional distributions superpose. If not treated specifically, this issue can create discontinuities in the likelihood. Therefore, we set the "problematic" resampled particles to the value of the closest regime.

In Section 4, we implement the smooth particle filter for the MS-SVL model. We observe that using 100,000 particles is still insufficient to obtain a smooth log-likelihood surface (as Fig. 2 shows), and implies large computation times. Therefore, we propose an approach in which the regime variable will receive special treatment.

3. Kim smooth particle filters

The following pseudo-code presents the main steps in our approach.

Step 0: Initialization
Given Θ, set $p(s_0;\Theta)$ to the regime ergodic probabilities and draw S distributions $\{h^{(i)}_{s_0,0}, w^{(i)}_{s_0,0}=\frac{1}{M}\}_{i=1:M}$ in $\mathcal{N}(h_0;\mu_{s_0},\frac{\sigma_\eta^2}{1-\phi^2})$.

for $t=1,\ldots,T$ **do**

Step 1: Given $p(s_{t-1}|y_{1:t-1};\Theta)$, predict regime probabilities:

$$p(s_t,s_{t-1}|y_{1:t-1};\Theta)=p(s_t|s_{t-1};\Theta)\times p(s_{t-1}|y_{1:t-1};\Theta).$$

Step 2: Given the S distributions $\{h^{(i)}_{s_{t-1},t-1},\frac{1}{M}\}_{i=1:M}$, draw current particles $\tilde{h}^{(i)}_{s_t,s_{t-1},t}$ in the proposal $q(h_t|h^{(i)}_{s_{t-1},t-1},y_t;\Theta_{s_t})$ with weights

$$\hat{w}^{(i)}_{s_t,s_{t-1},t}\propto\frac{1}{M}\times\frac{p(y_t|\tilde{h}^{(i)}_{s_t,s_{t-1},t};\Theta_{s_t})\times p(\tilde{h}^{(i)}_{s_t,s_{t-1},t}|h^{(i)}_{s_{t-1},t-1};\Theta_{s_t})}{q(\tilde{h}^{(i)}_{s_t,s_{t-1},t}|h^{(i)}_{s_{t-1},t-1},y_t;\Theta_{s_t})}.$$

Calculate the conditional density:

$$p(y_t|s_t,s_{t-1},y_{1:t-1};\Theta)\approx\sum_{i=1}^{M}\hat{w}^{(i)}_{s_t,s_{t-1},t}.$$

$\forall s_t\in\{1,\ldots,S\}$, build $\left\{\tilde{h}^{(i)}_{s_t,s_{t-1},t},\frac{p(s_t,s_{t-1}|y_{1:t-1};\Theta)\times\hat{w}^{(i)}_{s_t,s_{t-1},t}}{\sum_{i=1}^{M}\sum_{s_{t-1}=1}^{S}p(s_t,s_{t-1}|y_{1:t-1};\Theta)\times\hat{w}^{(i)}_{s_t,s_{t-1},t}}\right\}_{i=1:M,s_{t-1}=1:S}$ and smooth resample M particles to obtain the distribution $\{h^{(i)}_{s_t,t},\frac{1}{M}\}_{i=1:M}$.

Step 3: Calculate the marginal density:

$$p(y_t|y_{1:t-1};\Theta)=\sum_{s_t=1}^{S}\sum_{s_{t-1}=1}^{S}p(y_t|s_t,s_{t-1},y_{1:t-1};\Theta)\times p(s_t,s_{t-1}|y_{1:t-1};\Theta).$$

Step 4: Update regime probabilities:

$$p(s_t|y_{1:t};\Theta)=\frac{\sum_{s_{t-1}=1}^{S}p(y_t|s_t,s_{t-1},y_{1:t-1};\Theta)\times p(s_t,s_{t-1}|y_{1:t-1};\Theta)}{p(y_t|y_{1:t-1};\Theta)}.$$

end for

3.1. Presentation

The main idea is that we can use the Hamilton filter to infer the regimes. Thus, steps 1, 3, and 4 will constitute the outer layer of our filter dedicated to the regime probabilities. Given $p(s_{t-1}|y_{1:t-1};\Theta)$, and Θ, **step 1** predicts the regime

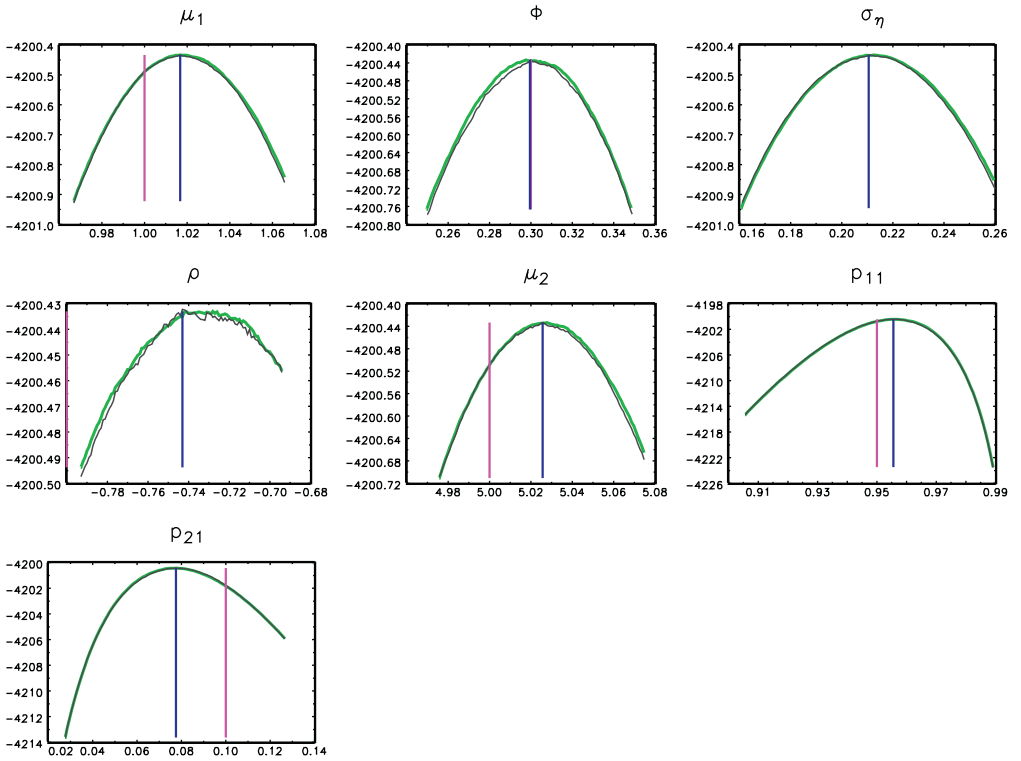


Fig. 2. Transversal cuts of the log-likelihood function by parameter (case 1): (MS-SVL model, smooth SIR, $M = 50{,}000$ (black line) and $M = 100{,}000$ (green line). Blue line: estimated value, pink line: true value. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

probabilities:

$$p(s_t, s_{t-1}|y_{1:t-1}; \Theta) = p(s_t|s_{t-1}; \Theta) \times p(s_{t-1}|y_{1:t-1}; \Theta). \tag{7}$$

It can reduce Monte-Carlo variation in the resulting estimator and improve numerical efficiency because we handle S regime probabilities rather than a large set of $M \times (S-1)$ particles approximating the regimes. Finally, it excludes the regime variable from the set of latent variables to resample, solving the limitations of the smooth particle filter evoked in the previous section.

However, a difficulty remains because the model is nonlinear conditional on the transition (s_{t-1}, s_t) and with an unobserved variable (namely, the log-volatility). To solve this issue, we use a particle filter to approximate the unobserved log-volatility and calculate an estimate of the likelihood conditional on $s_{t-1:t}$, respectively $p(h_t|s_t, s_{t-1}, y_{1:t-1}; \Theta)$ and $p(y_t|s_t, s_{t-1}, y_{1:t-1}; \Theta)$. It constitutes the inner layer of our filter, namely, **step 2**, and is mostly founded on the standard steps described in Section 2.2 conditioned by the regime variables.

Assume S sets of M particles denoted as $\{h^{(i)}_{s_{t-1},t-1}, w^{(i)}_{t-1} = \frac{1}{M}\}_{i=1:M}$, approximating $p(h_{t-1}|s_{t-1}, y_{1:t-1}; \Theta)$, the density of the log-volatility at time $t-1$ in regime s_{t-1} (each particle weight $w^{(i)}_{t-1}$ equals $\frac{1}{M}$, since we should use systematic resampling). To approximate $p(h_t|s_t, s_{t-1}, y_{1:t-1}; \Theta)$, we draw the current particles, $\tilde{h}^{(i)}_{s_t,s_{t-1},t}$, in the proposal $q(h_t|h^{(i)}_{s_{t-1},t-1}, s_t, y_t; \Theta)$. We discuss the choice of the proposal in the next subsection. We now have S^2 sets of M particles $\{\tilde{h}^{(i)}_{s_t,s_{t-1},t}\}_{i=1:M}$ for every (s_{t-1}, s_t) with (unormalized) weights:

$$\hat{w}^{(i)}_{s_t,s_{t-1},t} \propto w^{(i)}_{t-1} \frac{p(y_t|\tilde{h}^{(i)}_{s_t,s_{t-1},t}, s_t; \Theta) \times p(\tilde{h}^{(i)}_{s_t,s_{t-1},t}|h^{(i)}_{s_{t-1},t-1}, s_t; \Theta)}{q(\tilde{h}^{(i)}_{s_t,s_{t-1},t}|h^{(i)}_{s_{t-1},t-1}, s_t, y_t; \Theta)}. \tag{8}$$

An unbiased estimate of the conditional likelihood is provided by

$$p(y_t|s_t, s_{t-1}, y_{1:t-1}; \Theta) \approx \sum_{i=1}^{M} \hat{w}^{(i)}_{s_t,s_{t-1},t}. \tag{9}$$

Before implementing the third step of the algorithm, we should update the log-volatility distribution, which we do using univariate smooth resampling. It has two extra advantages: first, it prevents the algorithm from exploding after several iterations due to exponential complexity; second, we can implement a likelihood-based inference on parameters.

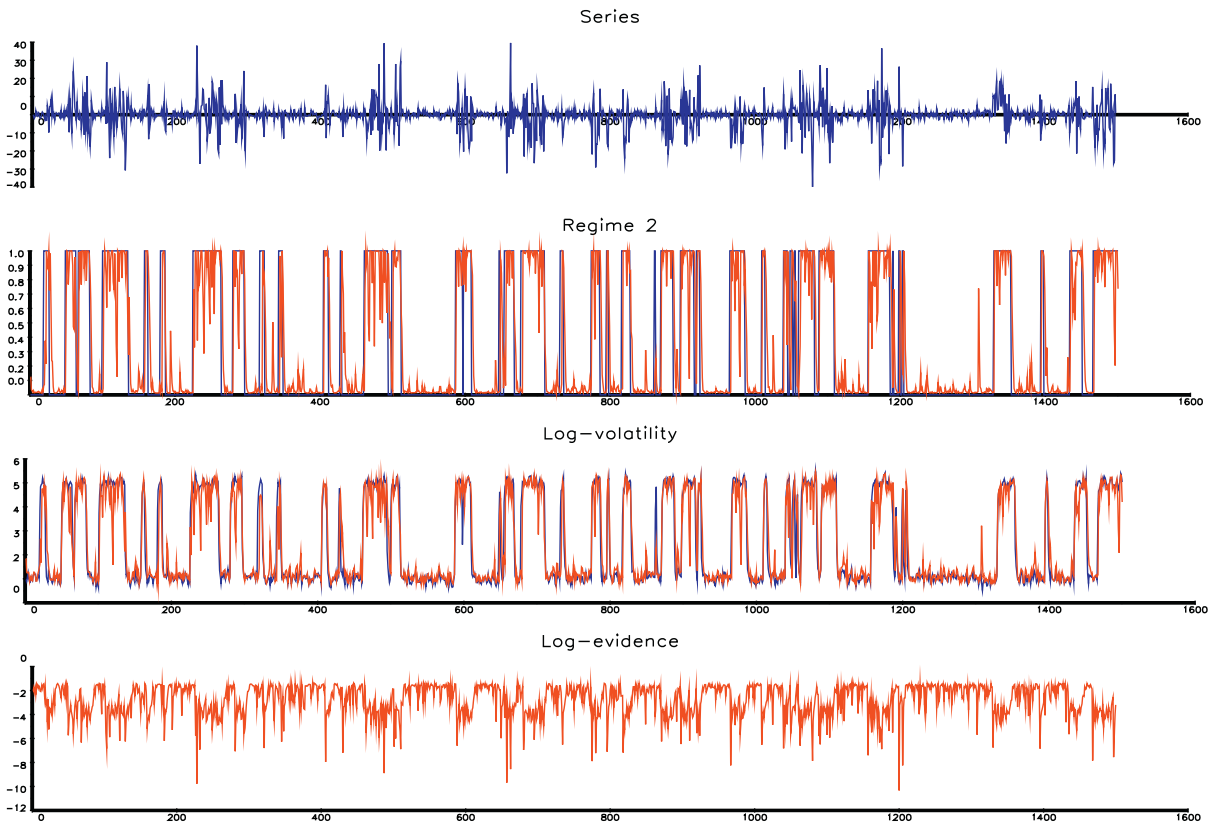


Fig. 3. Unobserved components provided by the LMDZ method (case 1): (MS-SVL model, $M = 75$). (a) Simulated returns (b) Regime 2 (true occurrence (in blue) vs. filtered probabilities (in red)) (c) Log-volatility (true (blue) vs. mean (red) estimates) (d) Estimated log-evidence (in red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

For every $s_t \in \{1, \ldots, S\}$, we consider S swarms of $M \times S$ particles with their unormalized weights $\{\tilde{h}^{(i)}_{s_t,s_{t-1},t}, p(s_t, s_{t-1}|y_{1:t-1}; \Theta) \times \hat{w}^{(i)}_{s_t,s_{t-1},t}\}_{i=1:M,s_{t-1}=1:S}$. After normalizing the weights, we smooth resample to obtain S swarms of M particles $\{h^{(i)}_{s_t,t}, w^{(i)}_t = \frac{1}{M}\}_{i=1:M}$ depending on s_t and approximating $p(h_t|s_t, y_{1:t}; \Theta)$. This step generalizes the collapsing technique employed by Kim (1994) for Gaussian distributions in a linear framework: S^2 particle swarms depending on s_t and s_{t-1} are replaced by S particle swarms depending only on s_t. It is also very similar to Chen and Liu (2000) and (Fearnhead and Clifford, 2003), who used non-smooth resampling schemes.

Using equations (7) and (9), **step 3** gives the marginal density, as in the standard approach:

$$p(y_t|y_{1:t-1}; \Theta) = \sum_{s_t=1}^{S} \sum_{s_{t-1}=1}^{S} p(y_t|s_t, s_{t-1}, y_{1:t-1}; \Theta) \times p(s_t, s_{t-1}|y_{1:t-1}; \Theta). \tag{10}$$

Step 4 updates the regime probabilities for the next time iteration based on the Bayes rule:

$$p(s_t|y_{1:t}; \Theta) = \frac{\sum_{s_{t-1}=1}^{S} p(y_t|s_t, s_{t-1}, y_{1:t-1}; \Theta) \times p(s_t, s_{t-1}|y_{1:t-1}; \Theta)}{p(y_t|y_{1:t-1}; \Theta)}. \tag{11}$$

3.2. Choosing the proposal: blind choice

In the spirit of the Bootstrap particle filter by Gordon et al. (1993), a first natural choice for the proposal is the transition distribution conditional on the current regime and past particles:

$$\begin{aligned} q(h_t|h^{(i)}_{s_{t-1},t-1}, s_t, y_t; \Theta) &= p(h_t|h^{(i)}_{s_{t-1},t-1}, s_t; \Theta) \\ &= p(h_t|h^{(i)}_{s_{t-1},t-1}; \Theta_{s_t}), \end{aligned} \tag{12}$$

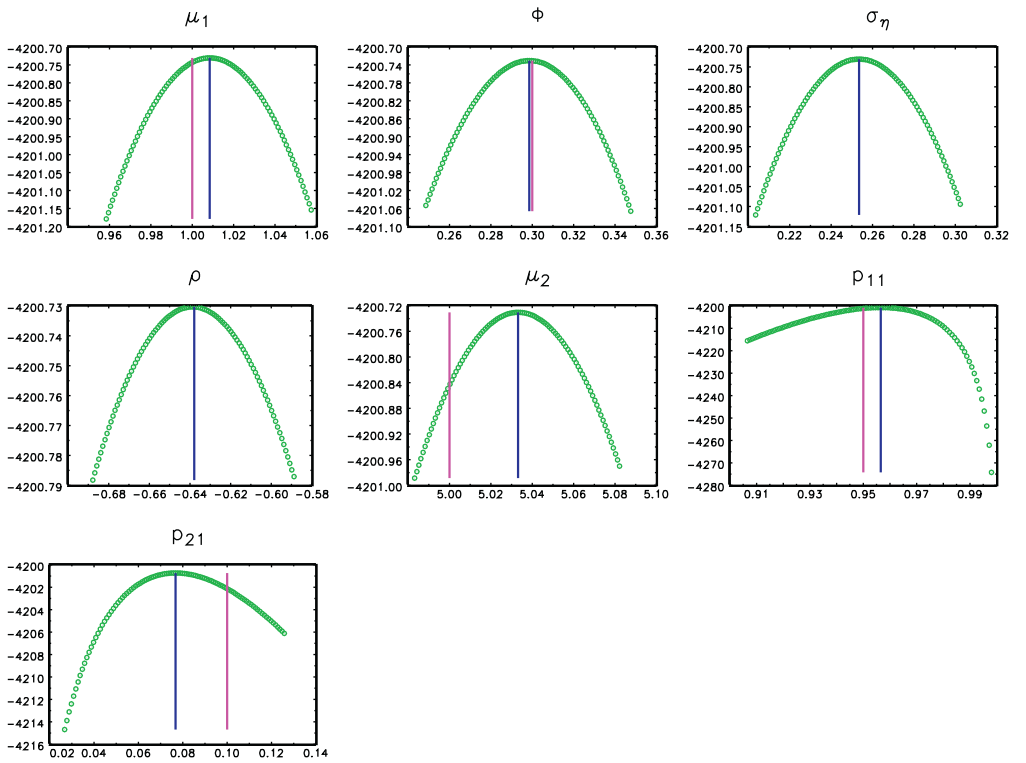


Fig. 4. Transversal cuts of the log-likelihood function by parameter (case 1): (MS-SVL model, LMDZ, $M = 75$). Blue line: estimated value, pink line: true value. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

with $\Theta_{s_t} = \{\mu_{s_t}, \phi, \sigma_\eta, \rho\}$, the subset of parameters in regime s_t. The associated unormalized weight simplifies

$$\begin{aligned}\hat{w}^{(i)}_{s_t,s_{t-1},t} &\propto w^{(i)}_{t-1} \frac{p(y_t|\tilde{h}^{(i)}_{s_t,s_{t-1},t};\Theta_{s_t}) \times p(\tilde{h}^{(i)}_{s_t,s_{t-1},t}|h^{(i)}_{s_{t-1},t-1};\Theta_{s_t})}{q(\tilde{h}^{(i)}_{s_t,s_{t-1},t}|h^{(i)}_{s_{t-1},t-1},y_t;\Theta_{s_t})} \\ &= \frac{1}{M} \times p(y_t|\tilde{h}^{(i)}_{s_t,s_{t-1},t};\Theta_{s_t}).\end{aligned} \tag{13}$$

This choice makes the procedure easy to implement and confers a natural interpretation to the particle weight, as it is proportional to its contribution to the current observation likelihood.

Using this proposal, the four steps above constitute the "Kim Bootstrap Smooth Particle Filter" (KBSPF hereafter). Appendix B-(2) summarizes the practical steps of this approach.

3.3. A more efficient proposal

The limit generally evoked in the literature is that $p(h_t|h^{(i)}_{s_{t-1},t-1};\Theta_{s_t})$ is a "blind" proposal because it does not depend on the current measurement variables y_t. To obtain a better proposal, we use the Gaussian *posterior* distribution from a nonlinear Kalman Filter (Kotecha and Djuric, 2003; Winschel and Kratzig, 2010, or Murray et al., 2013), which takes into account the current information y_t. We derive it by modifying only step 2, as follows.

Given the S conditional distributions $\{h^{(i)}_{s_{t-1},t-1}, \frac{1}{M}\}_{i=1:M}$ and the current regime s_t, we use the state-space representation to forecast the log-volatility h_t and y_t for each transition (s_{t-1}, s_t). It gives the particle swarms $\{h^{(i)}_{s_t,s_{t-1},t|t-1}\}_{i=1:M}$ and $\{y^{(i)}_{s_t,s_{t-1},t|t-1}\}_{i=1:M}$ (we use the standard notation for the Kalman filter from the literature, namely, $x_{t|t-1}$ is the prediction for variable x at time t conditional on the information available up to $t-1$):

$$\begin{aligned}h^{(i)}_{s_t,s_{t-1},t|t-1} &= f(h^{(i)}_{s_{t-1},t}, \xi^{(i)}_t;\Theta_{s_t}), \\ y^{(i)}_{s_t,s_{t-1},t|t-1} &= g(h^{(i)}_{s_t,s_{t-1},t|t-1}, \varepsilon^{(i)}_t;\Theta_{s_t}).\end{aligned}$$

$\{h^{(i)}_{s_t,s_{t-1},t|t-1}\}_{i=1:M}$ approximates the Kalman *prior* distribution, which is equivalent to the transition distribution $p(h_t|h^{(i)}_{s_{t-1},t-1};\Theta_{s_t})$ described in the previous subsection. As $\mathbb{E}[h_t|s_{t-1:t}, y_{1:t-1};\Theta]$ and $\mathbb{V}[h_t|s_{t-1:t}, y_{1:t-1};\Theta]$, the moments of

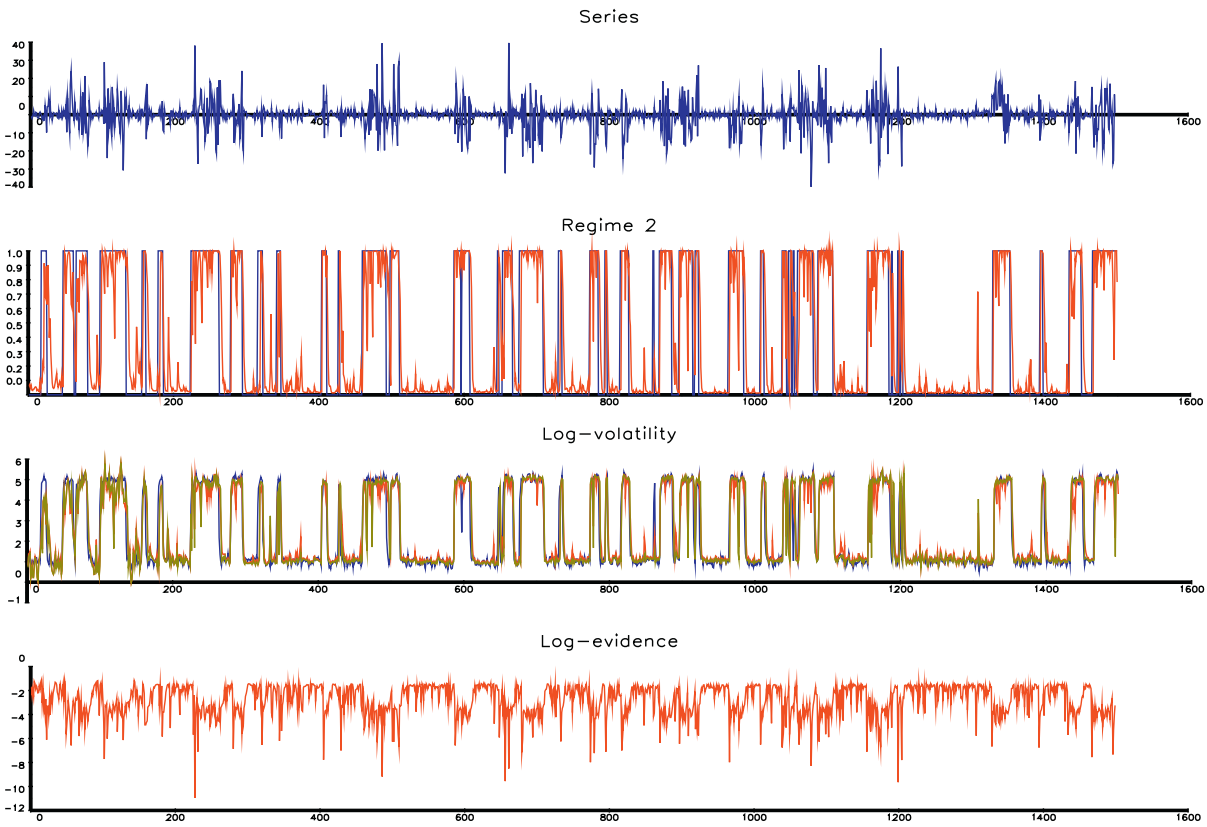


Fig. 5. Unobserved components provided by the online filter (case 1): (MS-SVL model, $\delta = 0.99, M = 100{,}000$). (a) Simulated returns (b) Regime 2 (true occurrence (in blue) vs. filtered probabilities (in red)) (c) Log-volatility (true (in blue) vs. mean (in red) and median (green) estimates) (d) Estimated log-evidence (in red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the *prior* distribution, $\mathbb{E}[y_t|s_{t-1:t}, y_{1:t-1}; \Theta]$, $\mathbb{V}[y_t|s_{t-1:t}, y_{1:t-1}; \Theta]$ and $\mathbb{C}ov[h_t, y_t|s_{t-1:t}, y_{1:t-1}; \Theta]$ cannot be obtained directly, as in a standard Kalman filter, we estimate them using the empirical first- and second-order moments of both unobserved and observed predicted distributions:

$$h_{s_t,s_{t-1},t|t-1} = \frac{1}{M}\sum_{i=1}^{M} h^{(i)}_{s_t,s_{t-1},t|t-1},$$

$$P_{h_{s_t,s_{t-1},t|t-1}} = \frac{1}{M}\sum_{i=1}^{M} (h^{(i)}_{s_t,s_{t-1},t|t-1} - h_{s_t,s_{t-1},t|t-1})^2,$$

$$y_{s_t,s_{t-1},t|t-1} = \frac{1}{M}\sum_{i=1}^{M} y^{(i)}_{s_t,s_{t-1},t|t-1},$$

$$P_{y_{s_t,s_{t-1},t|t-1}} = \frac{1}{M}\sum_{i=1}^{M} (y^{(i)}_{s_t,s_{t-1},t|t-1} - y_{s_t,s_{t-1},t|t-1})^2,$$

$$P_{y,h_{s_t,s_{t-1},t|t-1}} = \frac{1}{M}\sum_{i=1}^{M} (y^{(i)}_{s_t,s_{t-1},t|t-1} - y_{s_t,s_{t-1},t|t-1})(h^{(i)}_{s_t,s_{t-1},t|t-1} - h_{s_t,s_{t-1},t|t-1}),$$

to deduce the Kalman gain

$$K_{s_t,s_{t-1},t} = P_{y,h_{s_t,s_{t-1},t|t-1}} (P_{y_{s_t,s_{t-1},t|t-1}})^{-1}.$$

We then implement the Kalman updating step to estimate $\mathbb{E}[h_t|s_{t-1}, s_t, y_{1:t}; \Theta]$ and $\mathbb{V}[h_t|s_{t-1}, s_t, y_{1:t}; \Theta]$, the moments of the *posterior* distribution that takes into account the current information y_t:

$$h_{s_t,s_{t-1},t|t} = h_{s_t,s_{t-1},t|t-1} + K_{s_t,s_{t-1},t}(y_t - y_{s_t,s_{t-1},t|t-1}),$$

$$P_{h_{s_t,s_{t-1},t|t}} = P_{h_{s_t,s_{t-1},t|t-1}} - K_{s_t,s_{t-1},t} P_{y_{s_t,s_{t-1},t|t-1}} (K_{s_t,s_{t-1},t})'.$$

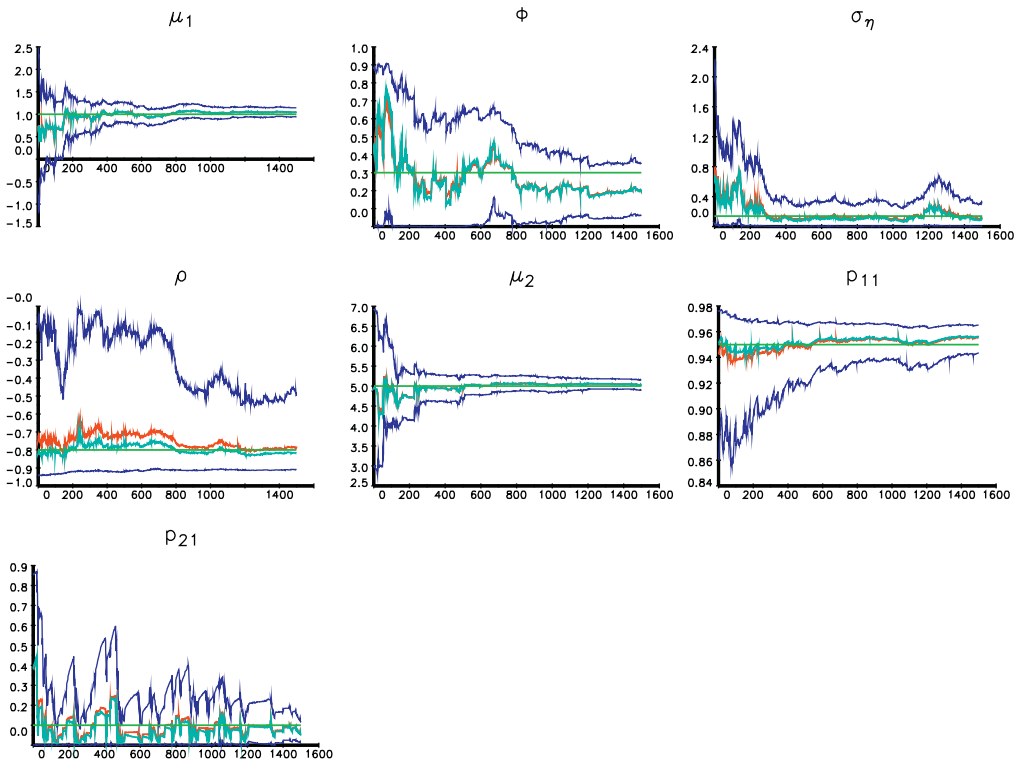


Fig. 6. Online estimation of parameters (case 1): (MS-SVL model, $\delta = 0.99, M = 100{,}000$). Red: parameter mean estimate, cyan: parameter median estimate, blue: confidence band at 95%, horizontal green line: true value. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

For each transition, under the Gaussian assumption, we draw the current particles $\{\tilde{h}^{(i)}_{s_t,s_{t-1},t}\}_{i=1:M}$ in the *posterior* distribution of the log-volatility $\mathcal{N}(h_t; h_{s_t,s_{t-1},t|t}, P_{h_{s_t,s_{t-1},t|t}})$ given by the nonlinear Kalman filter. From Eq. (8), the weights are:

$$\hat{w}^{(i)}_{s_t,s_{t-1},t} \propto \frac{1}{M} \times p(y_t|\tilde{h}^{(i)}_{s_t,s_{t-1},t}; \Theta_{s_t}) \frac{\mathcal{N}(\tilde{h}^{(i)}_{s_t,s_{t-1},t}; h_{s_t,s_{t-1},t|t-1}, P_{h_{s_t,s_{t-1},t|t-1}})}{\mathcal{N}(\tilde{h}^{(i)}_{s_t,s_{t-1},t}; h_{s_t,s_{t-1},t|t}, P_{h_{s_t,s_{t-1},t|t}})}.$$

We smooth resample as described in Section 3.2 to obtain S swarms of M particles $\{h^{(i)}_{s_t,t}, w^{(i)}_t = \frac{1}{M}\}_{i=1:M}$ only depending on regime s_t and approximating $p(h_t|s_t, y_{1:t}; \Theta)$.

Using this new step 2 with steps 1, 3, and 4 constitutes the "Kim Kalman Smooth Particle Filter" (2KSPF hereafter). Appendix B-(3) summarizes the practical steps of this approach.

3.4. Maximum likelihood

Since we use smooth resampling, we can implement direct maximum likelihood to estimate the model parameters:

$$\hat{\Theta}_T = \arg\max_{\Theta} \sum_{t=1}^{T} \log p(y_t|y_{1:t-1}; \Theta),$$

with the expression of the likelihood approximated by Eq. (10).

3.5. Smooth inference on regimes

One final advantage is that it easily provides smoothed inference on probabilities after the estimation and final filtering using the Kim backward smoother (see Kim and Nelson, 1999): $\forall t = T-1, \ldots, 1$:

$$p(s_t|y_{1:T}; \Theta) = \sum_{s_{t+1}=1}^{S} p(s_{t+1}|y_{1:T}; \Theta) p(s_t|y_{1:t}; \Theta) \frac{p(s_{t+1}|s_t; \Theta)}{p(s_{t+1}|y_{1:t}; \Theta)}.$$

This last remark is not specific to our approach and is possible for all competing methods by calculating the necessary probabilities at each date using the particle swarm. However, to our knowledge, no prior studies implement this method.

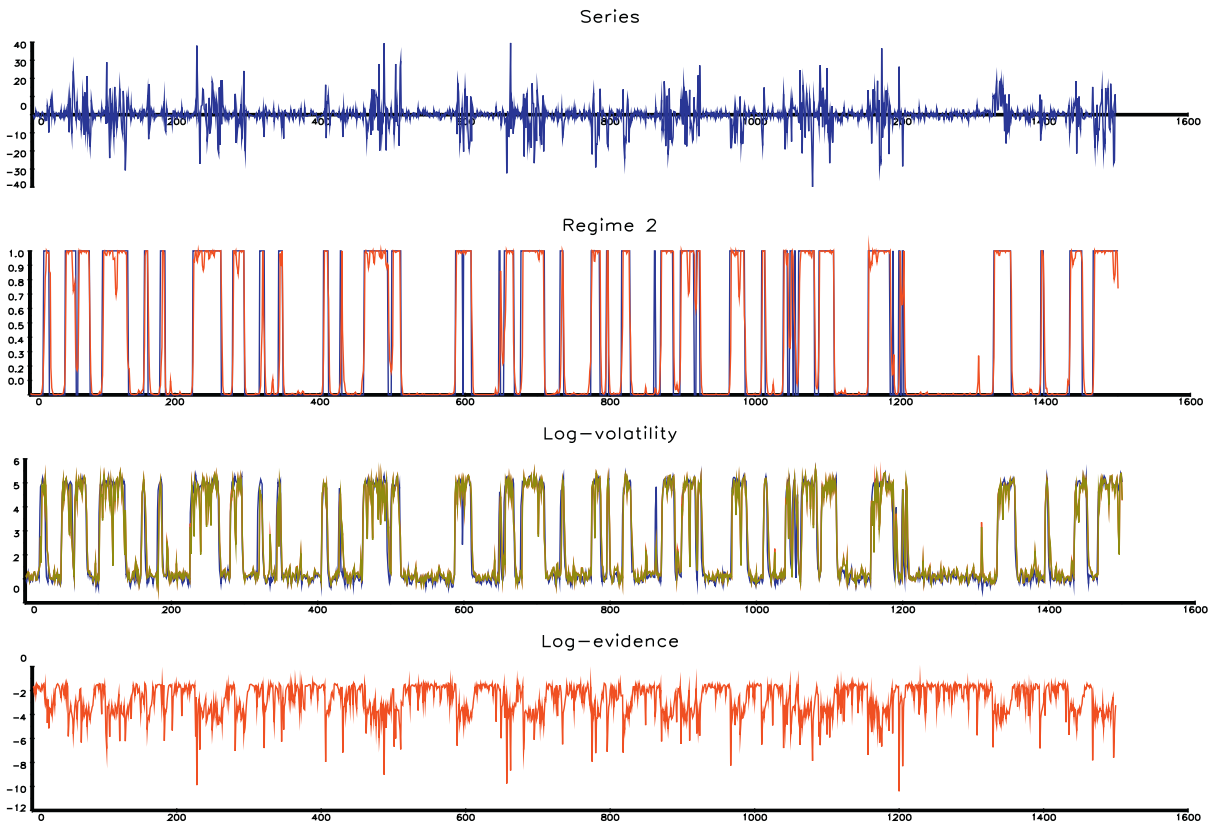


Fig. 7. Unobserved components provided by the KBSPF (case 1): (MS-SVL model, $M = 1{,}000$). (a) Simulated returns (b) Regime 2 (true occurrence (in blue) vs. smoothed probabilities (in red)) (c) Log-volatility (true (blue) vs. mean (red) and median (green) estimates) (d) Estimated log-evidence (in red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Simulation experiments

We propose two types of experiments to demonstrate the performance of our approach. First, we run an estimation exercise on a dataset simulated for a particular parameterization. This provides the first insight into the performance or limits of the tested methods, such as the inference on latent variables, likelihood smoothness, and computation times. Second, we run a Monte-Carlo experiment to assess the properties of the parameter estimators, the fit of the log-volatility, and the regime occurrence.

We use three methods as benchmarks for KBSPF and 2KSPF. The first is the smooth particle filter discussed in Section 2.2 (or smooth SIR hereafter). The second benchmark method is from Langrock et al. (2012) (LMDZ hereafter). We capture the regime probabilities with a Hamilton filter and approximate the continuous log-volatility using a fixed discrete grid that covers the entire support of the log-volatility. Each node of the grid is associated with a probability, and the probability distributions differ across regimes. Due to the leverage effect (see Eq. (6)), we simulate the state equation for each node of the grid conditional on the current and past regimes at each time to build a transition probability matrix, as in Tauchen (1986). Since this algorithm has the same structure as the filter in Kim (1994), we implement the collapsing technique to prevent algorithm explosion after several iterations. All calculations are based on standard summations and maximum likelihood can be implemented as usual (see Appendix B-(4) for an extensive description). The third benchmark method is the online filter proposed by Carvalho and Lopes (2007). To circumvent the estimation issue associated with standard particle filtering, these authors consider the unknown parameters as surrogate state variables and combine the auxiliary particle filter proposed by Pitt and Shephard (1999) with the (Liu and West, 2001) filter (see Appendix B-(5) for a full description). While we focus on the frequentist approach in this study, we use this online filter because it was derived specifically for the MS-SV model and is now used for a large set of models and applications (see, e.g., Bilio and Casarin (2010) or Bilio et al. (2010)). However, we know little about the properties of its estimators, which is a limitation frequently highlighted in the literature (Kantas et al., 2015). Therefore, it could be interesting to investigate this point with a Monte-Carlo experiment.

We implement the smooth SIR and the online filter with 50,000 and 100,000 particles, respectively. As is usually the case in studies on the online filter, we set $\delta = 0.99$, as Liu and West (2001) recommends. The parameters point estimate

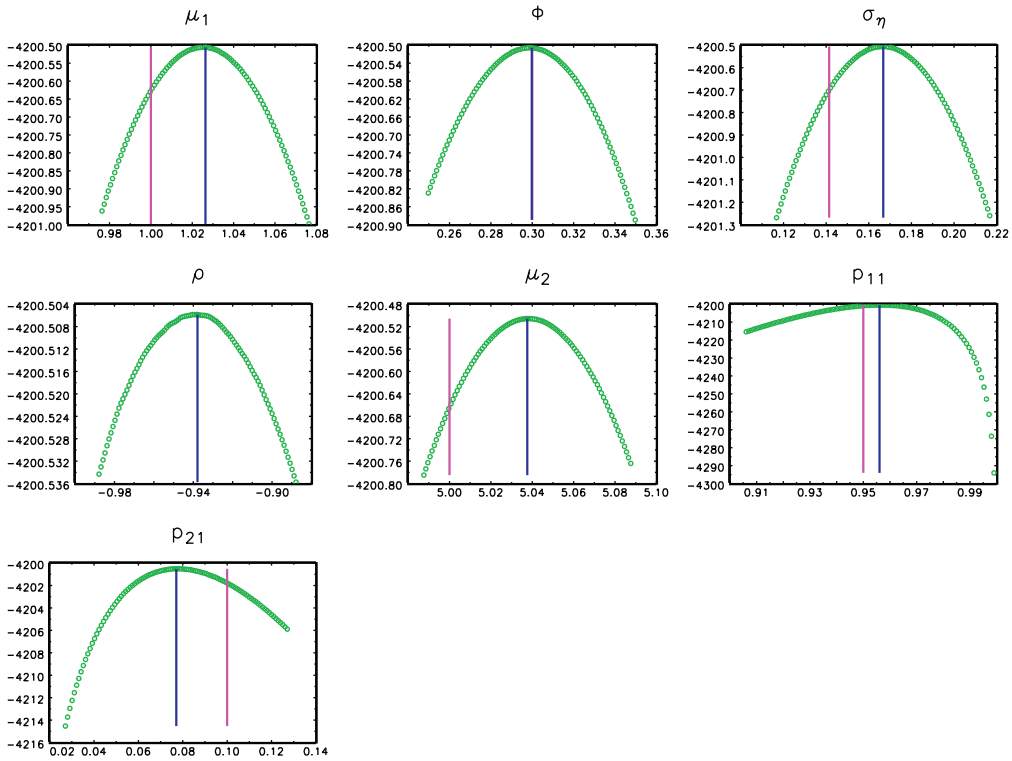


Fig. 8. Transversal cuts of the log-likelihood function by parameter (case 1): (MS-SVL model, KBSPF, $M = 1{,}000$). Blue line: estimated value, pink line: true value. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

is obtained as the mean of filtered particles for the last sample observation: $\hat{\Theta}_{online} = \frac{1}{M}\sum_{i=1}^{M}\Theta_T^{(i)}$. The LMDZ approach is implemented using a common grid of $M = 75$ values in each regime, as recommended by the authors (using a grid of length $M = 25$ does not change the results here). Finally, we implement KBSPF and 2KSPF with only 1,000 particles.

4.1. Experiment with a simulated dataset

We simulate a time-series of $T = 1{,}500$ observations with the MS-SVL model for any parameter values. We suppose $S = 2$ regimes with $s_t = 1$ and 2 being the low and high volatility regimes, respectively. We assess the performance of each method based on (*i*) the estimated parameters with regard to their true values and (*ii*) the fit of the latent variables with regard to their true value measured by the Root Mean Squared Error (RMSE, hereafter).

Figs. 1, 3, 5, 7, and 9 illustrate the fit for probabilities in regime 2, the log-volatility and the log-evidence for the smooth SIR, the LMDZ approach, the online filter, KBSPF and 2KSPF, respectively. Graphically, we can see that all methods capture log-volatility and regime occurrences well. The RMSEs show that all approaches deliver similar results for log-volatility (Table 1). However, using KBSPF or 2KSPF confers a non-negligible gain in accuracy for regime occurrence because we use smoothed probabilities. Table 1 also presents the parameter estimation results. All methods deliver very good results.

Fig. 2 displays the transversal cuts of the likelihood for each parameter (all other parameters being equal to the solution estimate). The solution provided by the smooth SIR with 50,000 particles is relatively close to the true vector of parameters. However, this result is slightly biased because transversal cuts are not completely smooth. Particularly, we do not obtain the mode for ρ, which seems closer to -0.73, due to the remaining discontinuities. Increasing the number of particles might counteract this issue. Fig. 2 also displays the transversal cuts calculated with 100,000 particles for the same solution. The result is more satisfying, except for ρ, which still exhibits discontinuities. Furthermore, the computation time to calculate one evaluation of the sample likelihood has also been multiplied by 2 (from 2 min 25 s to 4 min 55 s), even using some CPU parallelization (for particles and in the resampling step). Thus, this method is hardly implementable for estimation, even with important computational resources.

The LMDZ method delivers very similar results with a significant gain in terms of computation time (nearly 14 s for one likelihood evaluation). However, we notice that the approximation in this approach seems to worsen the inferences for σ_η and ρ.

The online filter is quite fast, even with a large sample size and a large number of particles because it requires only one pass over the sample to obtain results. Here, the complete estimation takes less than 13 min. However, while (i) the true

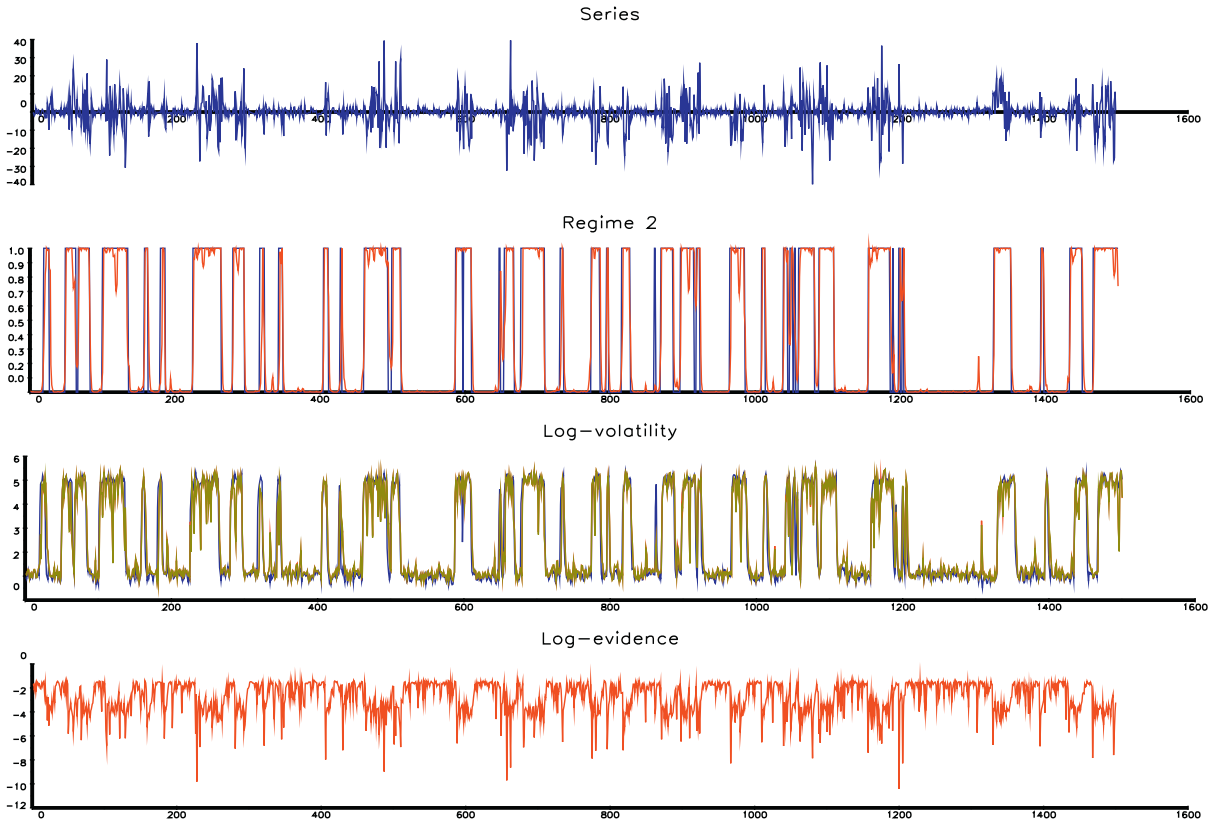


Fig. 9. Unobserved components provided by the 2KSPF (case 1): (MS-SVL model, $M = 1{,}000$). (a) Simulated returns (b) Regime 2 (true occurrence (in blue) vs. smoothed probabilities (in red)) (c) Log-volatility (true (blue) vs. mean (red) and median (green) estimates) (d) Estimated log-evidence (in red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Comparison of the estimation methods (MS-SVL model, benchmark sample (case 1), $T = 1{,}500$).

	DGP	Smooth SIR ($M = 5.10^4$)	LMDZ ($M = 75$)	Online ($\delta = 0.99, M = 10^5$)	KBSPF ($M = 1{,}000$)	2KSPF ($M = 1{,}000$)
	True value	Mode	Mode (Std-Err)	Mean [2.5%; 97.5%]	Mode (Std-Err)	Mode (Std-Err)
μ_1	1.00	1.0167	1.0113 (0.0575)	1.0475 [0.9491 ; 1.1457]	1.0264 (0.0410)	1.0249 (0.0473)
μ_2	5.00	5.0257	5.0179 (0.0765)	5.0311 [4.9037 ; 5.1550]	5.0375 (0.0762)	5.0298 (0.0742)
ϕ	0.30	0.2995	0.2944 (0.0696)	0.1983 [0.060 ; 0.3481]	0.2997 (0.0133)	0.2975 (0.0137)
σ_η	0.141	0.2106	0.2479 (0.0896)	0.1267 [0.0057 ; 0.3316]	0.1667 (0.0139)	0.1906 (0.0065)
ρ	−0.80	−0.7431	−0.6350 (0.2190)	−0.7910 [−0.9095 ; −0.4985]	−0.9378 (0.0598)	−0.8204 (0.0889)
p_{11}	0.95	0.9555	0.9565 (0.0073)	0.9555 [0.9432 ; 0.9650]	0.9560 (0.0068)	0.9570 (0.0071)
p_{21}	0.10	0.0776	0.0767 (0.0133)	0.0580 [0.0130 ; 0.1205]	0.0770 (0.0055)	0.0772 (0.0126)
Log-Likelihood		−4,200.4334	−4,200.7646	−4,219.8314	−4,200.5058	−4,200.2503
RMSE: log-vol.		0.7185	0.7196	0.7632	0.7172	0.7174
RMSE: regime 2		0.3017	0.3018	0.3123	0.2107	0.2107
CPU time[a]		2 min 25s	13.95s	12 min 48s	3.88s	4.95s

[a] It corresponds to the time required for one sample likelihood evaluation with a DELL XPS INTEL Core™i7-3770 CPU@3.40Ghz and 16 Go RAM. Standard-errors of parameters are provided into parenthesis.

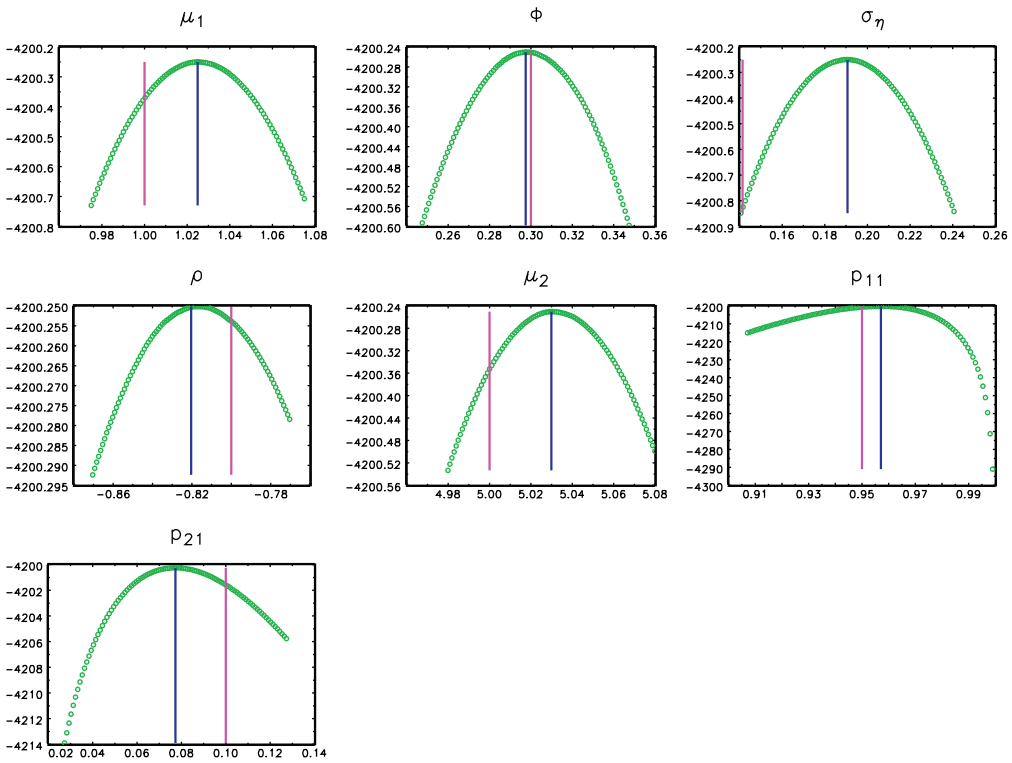


Fig. 10. Transversal cuts of the log-likelihood function by parameter (case 1): (MS-SVL model, 2KSPF, $M = 1{,}000$). Blue line: estimated value, pink line: true value. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

parameters are constant in the DGP, (ii) we start from the true values, and (iii) the number of particles is far greater than that in Carvalho and Lopes (2007), we observe that we can obtain a relatively inaccurate estimate (ϕ for instance) and/or the confidence band can be quite large (see Fig. 6 for ρ).

KBSPF delivers accurate parameter estimates but is slightly deceptive in the leverage effect (increasing the number of particles does not change this result), in line with the findings by Malik and Pitt (2011) for a model with no change in regime. Fig. 8 displays the transversal cuts of the likelihood from KBSPF for each parameter. We can see that smoothness is reached with only 1000 particles (the support for all transversal cut figures have the same length for ease of comparison). Consequently, KBSPF is considerably faster (nearly 4 s for one sample likelihood evaluation) because it requires far fewer particles to obtain acceptable results. 2KSPF preserves and outperforms the good performance of KBSPF when implemented in the same conditions. Using the nonlinear Kalman filter allows us to reintroduce (even imperfectly) information on the current observables in the proposal distribution and greatly improves the inference on the leverage effect without notably deteriorating the computation time (nearly 5 s for one sample likelihood evaluation). This result is important with regard to the two relatively strong assumptions made in this approach, which could be considered an important limit. First, we implement the Kalman updating step in our nonlinear framework, though it was formally derived for a linear framework. Second, the Gaussian assumption for the proposal distribution can be viewed as quite restrictive.

4.2. Monte-Carlo experiment

We now propose a Monte-Carlo experiment to assess the estimation performance of the competing approaches: LMDZ and the online filter vs. KBPSF and 2KSPF. We do not use smooth SIR here owing to its computational burden. Based on a parameterization generally encountered in the literature, we generate 500 artificial samples of $T = 1{,}500$ observations with the MS-SVL model. We use the same estimation conditions as above. The results are directly comparable since we implement the methods for the same samples using the same initial conditions. Fig. 11 displays the non-parametric estimates of the RMSEs' densities for regime occurrence and the log-volatility. We calculate the mean estimate, standard deviation, and root mean-square error for the parameters (Table 2) and measure the relative performance of the competing methods using the (Diebold and Mariano, 1995) test (Table 3). We summarize the results using non-parametric density estimation (Fig. 12).

Fig. 11 shows that LMDZ, KBPSF, and 2KSPF produce a similar fit for the log-volatility and clearly outperform the online filter. In terms of regime occurrence, KBPSF and 2KSPF provide a better fit than the two other methods due to the use of smoothed probabilities, which can be added to the LMDZ approach easily to improve its performances.

Table 2
Monte Carlo results: (MS-SVL model, 500 simulation experiments, $T = 1{,}500$).

Mean					
	DGP	LMDZ	Online	KBSPF	2KSPF
μ_1	−9.0000	−9.0232	−8.9045	**−9.0025**	−9.0114
μ_2	−7.0000	−6.9585	−6.7840	−6.9721	**−6.9768**
ϕ	0.9500	**0.9491**	0.9552	0.9484	0.9476
σ_η	0.1414	0.1314	0.1038	**0.1372**	0.1358
ρ	−0.8000	−0.8832	**−0.7840**	−0.8374	−0.8409
p_{11}	0.9900	0.9889	0.9879	**0.9897**	0.9891
p_{21}	0.0150	**0.0175**	0.0440	0.0211	0.0206
Standard-Error					
		LMDZ	Online	KBSPF	2KSPF
μ_1		**0.1262**	0.2697	0.1289	0.1357
μ_2		0.2036	0.4036	0.2029	**0.1886**
ϕ		0.0126	**0.0106**	0.0129	0.0127
σ_η		0.0242	0.0410	0.0243	**0.0226**
ρ		0.0939	0.0949	0.0921	**0.0848**
p_{11}		0.0062	0.0148	0.0061	**0.0051**
p_{21}		0.0121	0.0639	0.0108	**0.0093**
Root Mean Square Error					
		LMDZ	Online	KBSPF	2KSPF
μ_1		**0.1283**	0.2862	0.1289	0.1362
μ_2		0.2077	0.4578	0.2048	**0.1900**
ϕ		0.0126	**0.0118**	0.0130	0.0129
σ_η		0.0262	0.0556	0.0247	**0.0233**
ρ		0.1254	0.0963	0.0994	**0.0942**
p_{11}		0.0063	0.0150	0.0061	**0.0051**
p_{21}		0.0124	0.0702	0.0124	**0.0108**

The value in bold indicates the prevailing method.

Table 3
Diebold and Mariano statistics: (MS-SVL model, 500 simulation experiments, $T = 1,500$).

		Online		KBSPF		2KSPF	
		SEL	AEL	SEL	AEL	SEL	AEL
LMDZ	μ_1	−5.866	−7.157	−0.112	−0.352	−0.641	0.095
	μ_2	−6.049	−8.736	0.281	−0.075	1.650	1.675
	ϕ	1.371	0.233	−1.539	−1.334	−0.788	0.269
	σ_η	−18.024	−19.987	2.369	2.942	4.288	4.823
	ρ	5.275	11.479	9.967	9.959	10.495	11.739
	p_{11}	−1.587	2.714	0.885	0.098	4.890	7.385
	p_{21}	−6.480	−10.029	−0.009	−3.475	0.657	−0.081
Online	μ_1			5.883	7.155	5.460	7.032
	μ_2			6.163	8.703	6.415	9.434
	ϕ			−2.029	−1.020	−1.702	−0.049
	σ_η			18.744	21.645	19.456	23.063
	ρ			−0.501	−4.256	0.316	−1.825
	p_{11}			1.609	−2.733	1.697	−1.291
	p_{21}			6.479	9.522	6.527	9.998
KBSPF	μ_1					−0.652	0.500
	μ_2					2.383	2.793
	ϕ					0.624	2.147
	σ_η					3.165	3.037
	ρ					2.235	3.924
	p_{11}					4.618	8.089
	p_{21}					3.436	5.904

This table displays the t-stat for testing the nullity of the loss function mean (SEL stands for square error loss and AEL for absolute error loss) from two competing methods (the test is bilateral). Negative (positive) statistics means that the row method out- (under-) performs the column method.

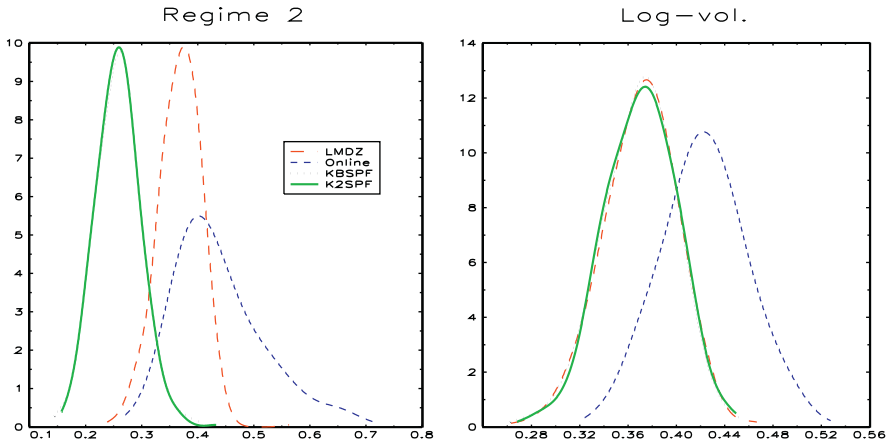


Fig. 11. RMSEs of latent variables: non-parametric density estimates with a Gaussian kernel, optimal bandwidth, MS-SVL model, 500 simulation experiments.

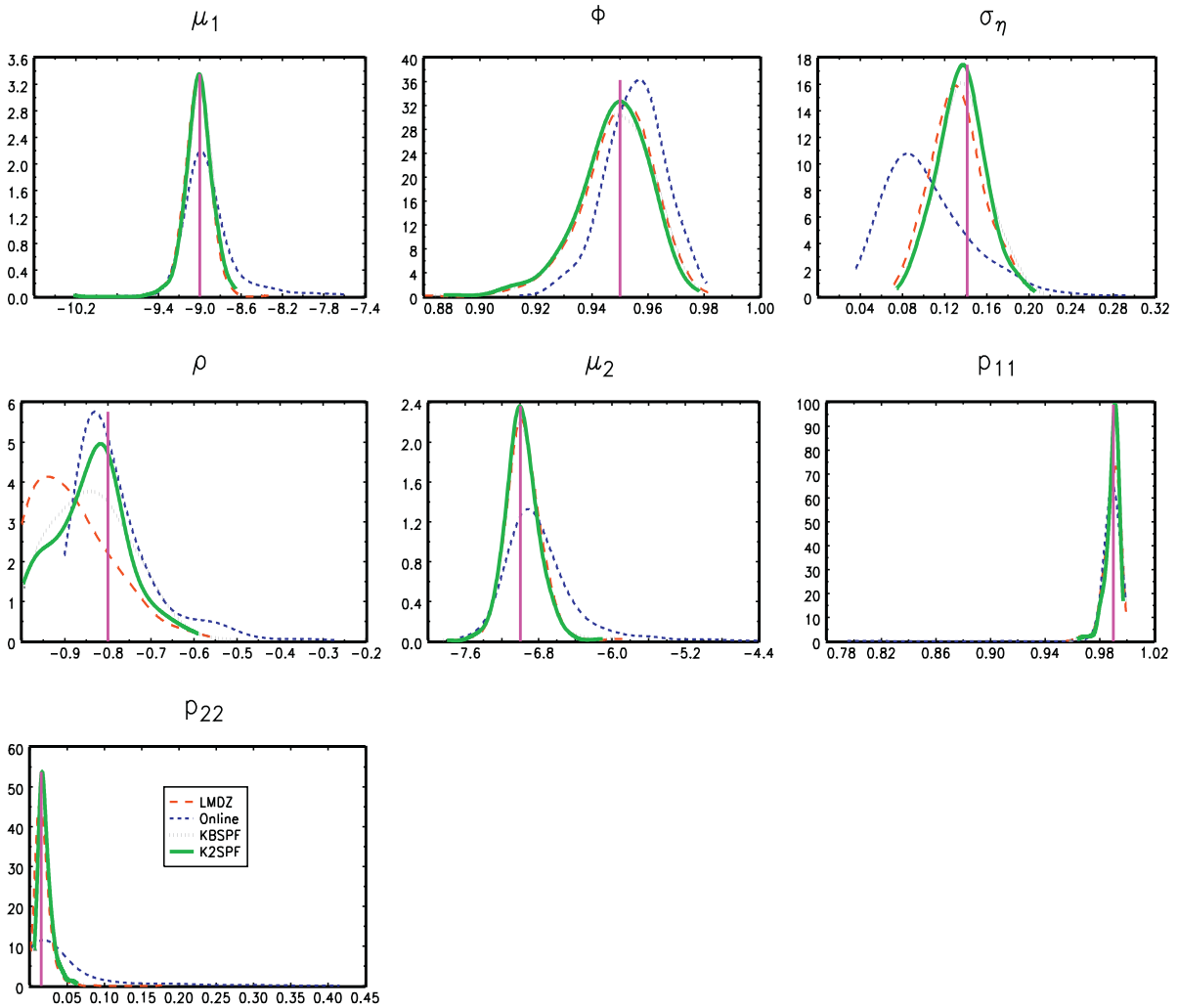


Fig. 12. Non-parametric density estimates for parameters: Gaussian kernel, optimal bandwidth, MS-SVL model, 500 simulation experiments. The pink vertical line indicates the true value. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Using our approach, and particularly 2KSPF, improves the quality of the estimates relative to LMDZ and the online filter. RMSEs indicate more accurate estimates in all cases, except for μ_1 and ϕ. Statistically, the Diebold and Mariano (1995) tests confirm that 2KSPF outperforms or is at least equivalent to all competing methods at 5%, whether we use the squared or absolute error criterion. Our approach clearly outperforms the LMDZ, particularly in the innovation variance, the leverage parameter ρ, and the transition probability p_{11}. The online filter is dominated in nearly all parameters, except for ϕ, ρ, and p_{11}. 2KSPF also outperforms KBSPF for nearly all parameters at 5%.

5. Application to real data

We now implement a comparative approach for a recent period (January 2006 to June 2015) of the S&P500 and NASDAQ for the USA and CAC40 for France, on a daily basis. We evaluate the volatility in the period, including the financial crisis (2007–2009) and the sovereign debt crisis (2011), for a three-regime representation *a la* (So et al., 1998). Therefore, we estimate the model for $S = 1$, $S = 2$, and $S = 3$ regimes, respectively. We present the estimation results using KBSPF in Table 4 (the estimation results with 2KSPF are quantitatively similar and not reproduced here. They are available upon request). Table 5 summarizes the regime properties of each model. Figs. 13 and 14 present the results for volatility and regime occurrence. The figures for other regime specifications are available in web Appendix B.

Table 4
Estimation results of MS-SVL using KBSPF.

	S&P500 01/03/2006-06/18/2015			NASDAQ 01/03/2006-06/17/2015			CAC40 01/03/2006-06/18/2015		
	$S=1$	$S=2$	$S=3$	$S=1$	$S=2$	$S=3$	$S=1$	$S=2$	$S=3$
μ_1	−9.197	−9.508	−9.862	−8.883	−9.211	−9.416	−8.876	−9.205	−9.294
	(0.597)	(0.062)	(0.047)	(0.091)	(0.061)	(0.047)	(0.078)	(0.065)	(0.052)
μ_2	–	−7.825	−8.892	–	−7.607	−8.595	–	−8.122	−8.550
		(0.083)	(0.086)		(0.119)	(0.099)		(0.098)	(0.139)
μ_3	–	–	−7.267	–	–	−7.147	–	–	−7.348
			(0.151)			(0.150)			(0.271)
ϕ	0.968	0.951	0.917	0.966	0.936	0.908	0.970	0.951	0.946
	(0.036)	(0.004)	(0.008)	(0.004)	(0.007)	(0.011)	(0.003)	(0.004)	(0.006)
σ_η	0.258	0.259	0.279	0.224	0.231	0.225	0.218	0.238	0.227
	(0.020)	(0.011)	(0.017)	(0.016)	(0.016)	(0.022)	(0.014)	(0.012)	(0.015)
ρ	−0.796	−0.875	−0.973	−0.775	−0.871	−0.976	−0.852	−0.927	−0.982
	(0.143)	(0.010)	(0.019)	(0.045)	(0.028)	(0.034)	(0.033)	(0.012)	(0.019)
p_{11}	–	0.999	0.996	–	0.998	0.996	–	0.997	0.996
		(0.000)	(0.001)		(0.001)	(0.002)		(0.002)	(0.002)
p_{21}	–	0.021	0.007	–	0.021	0.006	–	0.012	0.010
		(0.007)	(0.004)		(0.009)	(0.006)		(0.005)	(0.007)
p_{31}	–	–	0.012	–	–	0.013	–	–	0.000
			(0.021)			(0.021)			(0.024)
p_{12}	–	–	0.001	–	–	0.001	–	–	0.000
			(0.001)			(0.001)			(0.004)
p_{22}	–	–	0.987	–	–	0.987	–	–	0.981
			(0.006)			(0.008)			(0.009)
p_{32}	–	–	0.021	–	–	0.020	–	–	0.038
			(0.020)			(0.022)			(0.031)
T	2,379	2,379	2,379	2,378	2,378	2,378	2,417	2,417	2,417
k	4	7	12	4	7	12	4	7	12
M	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
Log-lik.	7721.64	7730.26	7737.81	7357.90	7372.16	7385.18	7230.12	7239.57	7245.60
AIC	−6.488	−6.493	−6.495	−6.185	−6.194	−6.201	−5.979	−5.985	−5.986

Standard-errors of parameters are provided into parenthesis.

Table 5
Summary statistics on three-regime models.

	S&P500			NASDAQ			CAC40		
P	$\begin{pmatrix} 0.99676 & 0.00686 & 0.01215 \\ 0.00131 & 0.98715 & 0.02121 \\ 0.00193 & 0.00599 & 0.96665 \end{pmatrix}$			$\begin{pmatrix} 0.99649 & 0.00605 & 0.01388 \\ 0.00117 & 0.98729 & 0.02028 \\ 0.00234 & 0.00667 & 0.96584 \end{pmatrix}$			$\begin{pmatrix} 0.99617 & 0.01089 & 0.00004 \\ 0.00082 & 0.98188 & 0.03890 \\ 0.00301 & 0.00723 & 0.96106 \end{pmatrix}$		
	$s_t=1$	$s_t=2$	$s_t=3$	$s_t=1$	$s_t=2$	$s_t=3$	$s_t=1$	$s_t=2$	$s_t=3$
Erg. prob.	0.72	0.20	0.08	0.70	0.21	0.09	0.67	0.23	0.10
Exp. dur.	308.35	77.83	29.98	284.89	78.66	29.27	261.00	55.19	25.68
# obs.	1,341	786	252	1,527	613	238	1,497	651	269
% obs.	56	33	10	64	26	10	62	27	11

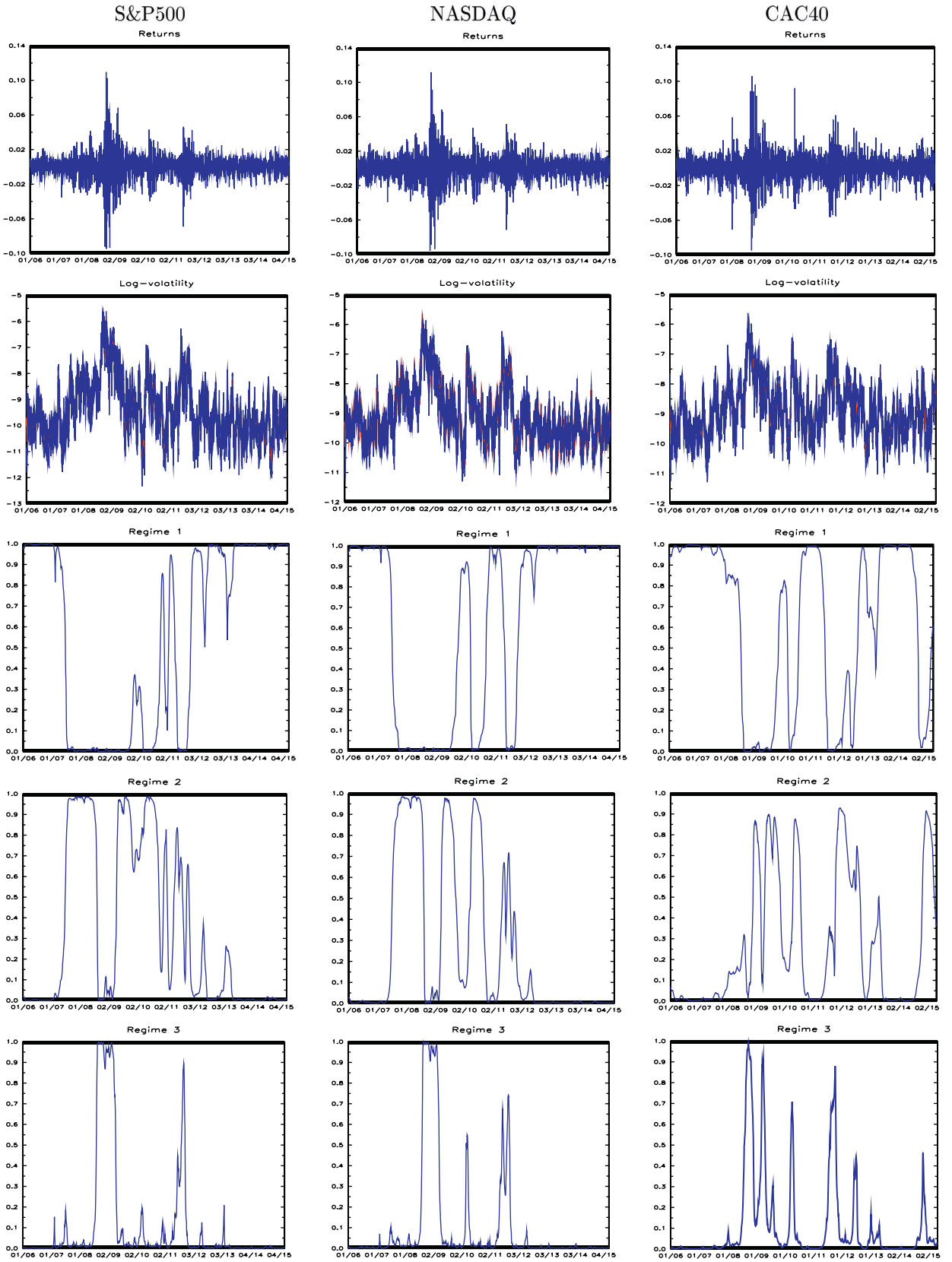


Fig. 13. Estimation results (from 01/03/2006-06/17/2015) using KBSPF: (a) Daily returns (b) Filtered log-volatility (mean and 95% confidence bands) (c)-(d)-(e) Regimes 1,2 and 3 (smoothed probabilities).

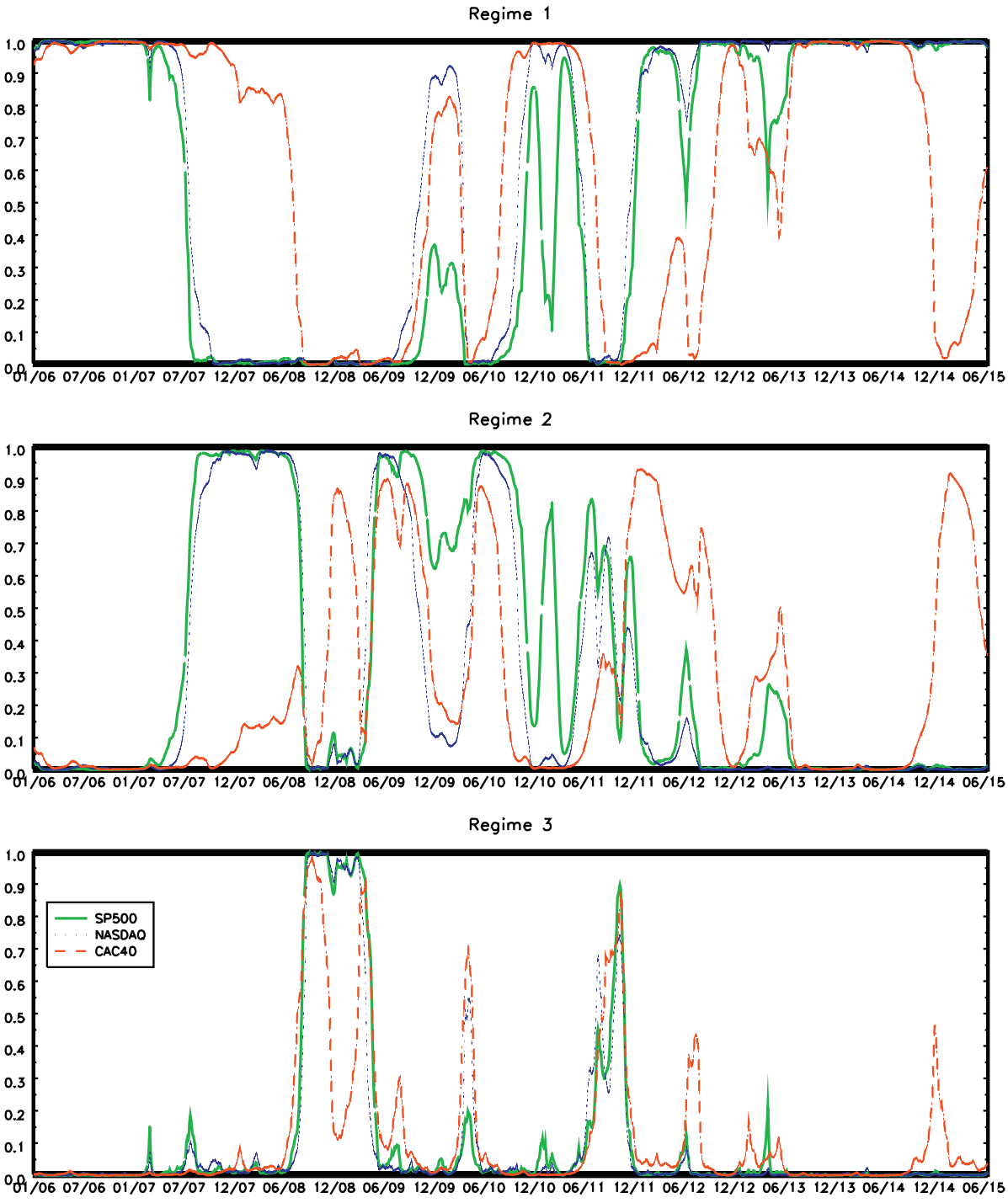


Fig. 14. Regimes 1,2 and 3 (smoothed probabilities): S&P500, NASDAQ and CAC40.

The AIC criterion is in favor of the model with three regimes, particularly for the NASDAQ, and to a lesser extent for the CAC40 and the S&P500. The estimation easily identifies three regimes with distinct levels of volatility. The persistence parameter ϕ decreases with the number of regimes, which is coherent with our previous findings and the type of values found in Malik and Pitt (2011). However, we do not obtain results similar to Hamilton and Susmel (1994), Cai (1994), or So et al. (1998) (around 0.48) with their approaches for previous periods. Unsurprisingly, we find a significant leverage effect ρ for all indices. We observe that its magnitude increases in absolute value with the number of regimes. Two reasons may

be invoked. On the one hand, the estimation may be poor in the model with only one regime because it neglects clusters of volatility. On the other hand, it could be a symptom of a misspecification of the model with three regimes, suggesting a regime-dependent leverage effect, as in Kim (2016).

The three series show very similar volatility profiles and regimes, particularly for the NASDAQ and S&P500, and very close properties in terms of expected duration and ergodic probabilities (see Table 5). The diagonal entries of the transition matrix p_{11}, p_{22}, and p_{33} are close to unity and decreasing (around 0.99, 0.98, and 0.96, respectively), implying that the regimes are persistent and the expected duration decreases with the level of volatility. The low-volatility regime prevails over the period (more than a year on average). On average, regimes 2 and 3 last about 78 and 30 days, respectively, in the US, while they last 55 and 25 days, respectively, in France. The estimated transition matrices also show the possibility of jumping directly from the low- to the high-volatility regime without necessarily visiting the medium one ($p_{13} > p_{12}$). Inversely, returning to the low volatility regime by passing in regime 2 ($p_{32} > p_{31}$) is more likely. These two outcomes are particularly true for the CAC40 index, for which p_{12} and p_{31} are very close to 0.

Fig. 13 shows that the model captures the 2007–2009 and 2011 crisis very well. The three indices show remarkable synchronicity in visiting the high-volatility regime (Fig. 14), which is associated with the peak events in both crisis. Starting from the low-volatility regime, the S&P500 and NASDAQ both switch to the medium-volatility regime in the first part of the crisis (from July 2007 to August 2008), corresponding to the sub-prime mortgage episode. France is not concerned and the CAC40 index remains in the low-volatility regime. The second phase (from September 2008 to May 2009) corresponds to the financial crisis and belongs to the high-volatility regime. The crisis became global and systemic for the banking system. France enters the crisis in the high-volatility regime in August 2008, without being in the medium-volatility regime. The US and France enter the Great Recession at the end of 2008 and at the beginning of 2009, respectively. In the wake of the measures implemented to sustain the banking system and other sectors, the return to the low-volatility regime is progressive (until the third quarter of 2010), through the intermediate volatility regime, closing out the Great Recession and giving way to the current period of stagnation.

The model also detects the sovereign debt crisis very well (from June to November 2011), which is also associated with the high-volatility regime. Afterwards, the two US indices return rapidly to the low-volatility regime by the end of 2011, while the CAC40 index remains in the medium-volatility regime during the first half of 2012, indicating the greater exposure of French banks to Spanish, Italian, and Greek debt. While the S&P500 and NASDAQ remain in the low-volatility regime from mid-2013 to mid-2015, the CAC40 index returns to the medium-volatility regime at the beginning of 2015, probably due to the increasing probability of Greece's recurrent default and the possibility of Grexit from the Euro zone.

6. Conclusion

This study proposes a new particle filtering approach for the stochastic volatility model with Markov-switching using maximum likelihood.

Our approach is faster than the benchmark approaches are because it requires a small number of particles to obtain accurate results. We demonstrate that it provides a very good fit for both volatility and regime occurrence because our probabilities are smoothed; that is, they incorporate all of the sample information. We also show that using a nonlinear Kalman filter as a proposal significantly improves the parameter estimations.

Acknowledgments

I thank the participants of the EPEE seminar (university Evry Val d'Essonne), the GAINS seminar (Le Mans University), the IRA seminar (Le Mans University), the workshop Filters (Le Mans University), the 4th workshop ISFA-IRA (Lyon), the Bayes in Paris Seminar (university Paris-Dauphine and ENSAE), the Computational and Financial Econometrics Conference (Pisa) and the PANORISK Conference: Measurement, Valuation & Modeling of Finance Risks (ESSCA Angers) for their remarks and comments. I thank more particularly Stephane Adjemian, Nicolas Chopin, John Maheu, Jonathan Wright, the two anonymous referees and the Associate Editor for their very constructive remarks and advice which have led to significant improvement of the paper. All remaining errors are my own. The author gratefully acknowledges financial support from the Région Pays de la Loire (France) through the grant PANORisk.

Appendix A. Univariate smooth resampling

Malik and Pitt (2011) propose a resampling method for a single state variable (univariate smooth resampling) to produce a likelihood estimator continuous only in the deep parameters of the model. Parameter inference is then possible through simulated maximum likelihood using particle filters.

Suppose we dispose of $\{\tilde{x}_t^{(i)}, \tilde{w}_t^{(i)}\}_{i=1:M}$, namely, the particle swarm for the current state variables drawn in the proposal distribution and their normalized weights. We define the weighted empirical distribution function as follows:

$$\hat{F}_M(x_t) = \sum_{i=1}^{M} \tilde{w}_t^{(i)} I(x_t - \tilde{x}_t^{(i)}),$$

where the $\{\tilde{x}_t^{(i)}\}_{i=1:M}$ are sorted in ascending order, the $\{\tilde{w}_t^{(i)}\}_{i=1:M}$ sum to unity, and $I(z)$ is an indicator function taking the value 1 if $z>0$ and 0 otherwise. The idea is now to approximate this function by

$$\tilde{F}_M(x_t) = \sum_{i=0}^{M} \tilde{\pi}_t^{(i)} G_i\left(\frac{x_t - \tilde{x}_t^{(i)}}{\tilde{x}_t^{(i+1)} - \tilde{x}_t^{(i)}}\right),$$

with $\tilde{x}_t^{(0)} = -\infty$, $\tilde{x}_t^{(M+1)} = +\infty$, $G_0(z) = G_M(z) = z$, $\tilde{\pi}_t^{(0)} = \frac{\tilde{w}_t^{(1)}}{2}$, $\tilde{\pi}_t^{(M)} = \frac{\tilde{w}_t^{(M)}}{2}$, and $\tilde{\pi}_t^{(i)} = \frac{\tilde{w}_t^{(i+1)} + \tilde{w}_t^{(i)}}{2}$ for $i = 1, \ldots, M-1$.

$\tilde{F}_M(.)$ is then a *continuous* cumulative distribution function that interpolates the *discrete* cumulative distribution $\hat{F}_M(.)$ by passing through the mid-points of each step. We then sample from the continuous density by (1) selecting region i with probability $\tilde{\pi}_t^{(i)}$ and (2) sampling within this region from $g_i(.)$, the density corresponding to $G_i(.)$. Following Malik and Pitt (2011), we choose the uniform distribution, so $G_i(z) = z$ for $i = 1, \ldots, M-1$. As in traditional resampling, we perform a single draw u in $U_{[0,\,1]}$ from which we calculate the sorted uniform draws as follows: $u^{(j)} = \frac{(j-1)+u}{M}$ for $j = 1, \ldots, M$.

In practice, we can implement univariate smooth resampling as follows. Define the continuous approximation of the cumulative density distribution $\bar{\pi}_t^{(i)} = \sum_{k=0}^{i} \tilde{\pi}_t^{(k)} = \bar{\pi}_t^{(i-1)} + \tilde{\pi}_t^{(i)}$ and the probability interval corresponding to region i as $[\bar{\pi}_t^{(i-1)}, \bar{\pi}_t^{(i)}[$.

- If $u^{(j)} < \bar{\pi}^{(0)}$, then the new particle is defined as $x_t^{(i)} = \tilde{x}_t^{(1)}$.
- If $u^{(j)} > \bar{\pi}^{(M)}$, then the new particle is defined as $x_t^{(i)} = \tilde{x}_t^{(M)}$.
- If $u^{(j)} \in [\bar{\pi}_t^{(i-1)}, \bar{\pi}_t^{(i)}[$, then we rescale $u^{(j)}$ with

$$u^{*(j)} = \frac{u^{(j)} - \bar{\pi}_t^{(i-1)}}{\bar{\pi}_t^{(i)} - \bar{\pi}_t^{(i-1)}},$$

and draw the new particle

$$x_t^{(i)} = \tilde{x}_t^{(i)} + (\tilde{x}_t^{(i+1)} - \tilde{x}_t^{(i)}) \times u^{*(j)}.$$

Appendix B. Filters algorithms

Algorithm 1 The smooth SIR.

Step 0: Initialization

Given Θ, draw $\{s_0^{(i)}\}_{i=1:M}$ in the regime ergodic probabilities distribution $p(s_0;\Theta)$; draw $\{h_0^{(i)}\}_{i=1:M}$ in the S-Gaussian mixture $\sum_{s_0=1}^{S} p(s_0;\Theta) \times \mathcal{N}(h_0; \mu_{s_0}, \frac{\sigma_\eta^2}{1-\phi^2})$. Set all weights to $w_0^{(i)} = \frac{1}{M}$.

for $t = 1, \ldots, T$ **do**

Step 1: Draw current particles in the transition densities:

$\tilde{s}_t^{(i)}$ from $p(s_t | s_{t-1}^{(i)};\Theta)$,

$\tilde{h}_t^{(i)}$ from $p(h_t | h_{t-1}^{(i)}, \tilde{s}_t^{(i)};\Theta)$.

Step 2: Calculate the unormalized weights:

$$\hat{w}_t^{(i)} \propto \frac{1}{M} \times p(y_t | \tilde{h}_t^{(i)}, \tilde{s}_t^{(i)};\Theta).$$

Compute the evidence as $\sum_{i=1}^{M} \hat{w}_t^{(i)}$ and normalize $\tilde{w}_t^{(i)} = \frac{\hat{w}_t^{(i)}}{\sum_{i=1}^{M} \hat{w}_t^{(i)}}$.

Step 3: Sort $\{\tilde{h}_t^{(i)}, \tilde{s}_t^{(i)}, \tilde{w}_t^{(i)}\}_{i=1:M}$ as regards $\tilde{h}_t^{(i)}$ and smooth resample to obtain $\{h_t^{(i)}, s_t^{(i)}, w_t^{(i)} = \frac{1}{M}\}_{i=1:M}$.

Step 4: If $s_t^{(i)} \neq \{0, 1\}$, set $s_t^{(i)}$ to the value of the closest regime.

end for

Algorithm 2 Kim Bootstrap Smooth Particle Filter (KBSPF).

Step 0: Initialization
Given Θ, set $p(s_0;\Theta)$ to the regime ergodic probabilities and draw S distributions $\{h^{(i)}_{s_0,0}, w^{(i)}_{s_0,0}=\frac{1}{M}\}_{i=1:M}$ in $\mathcal{N}(h_0;\mu_{s_0},\frac{\sigma^2_\eta}{1-\phi^2})$.

for $t=1,\ldots,T$ **do**

Step 1: Given $p(s_{t-1}|y_{1:t-1};\Theta)$, predict regime probabilities:

$$p(s_t,s_{t-1}|y_{1:t-1};\Theta)=p(s_t|s_{t-1};\Theta)\times p(s_{t-1}|y_{1:t-1};\Theta).$$

Step 2-1: Given the S distributions $\{h^{(i)}_{s_{t-1},t-1},\frac{1}{M}\}_{i=1:M}$, draw current particles $\tilde{h}^{(i)}_{s_t,s_{t-1},t}$ in the proposal $p(h_t|h^{(i)}_{s_{t-1},t-1};\Theta_{s_t})$ with weights

$$\hat{w}^{(i)}_{s_t,s_{t-1},t}\propto\frac{1}{M}\times p(y_t|\tilde{h}^{(i)}_{s_t,s_{t-1},t};\Theta_{s_t}).$$

Step 2-2: Calculate the conditional density:

$$p(y_t|s_t,s_{t-1},y_{1:t-1};\Theta)\approx\sum_{i=1}^{M}\hat{w}^{(i)}_{s_t,s_{t-1},t}.$$

Step 2–3: $\forall s_t\in\{1,\ldots,S\}$, build $\left\{\tilde{h}^{(i)}_{s_t,s_{t-1},t},\frac{p(s_t,s_{t-1}|y_{1:t-1};\Theta)\times\hat{w}^{(i)}_{s_t,s_{t-1},t}}{\sum_{i=1}^{M}\sum_{s_{t-1}=1}^{S}p(s_t,s_{t-1}|y_{1:t-1};\Theta)\times\hat{w}^{(i)}_{s_t,s_{t-1},t}}\right\}_{i=1:M,s_{t-1}=1:S}$ and smooth resample M particles to obtain the distribution $\{h^{(i)}_{s_t,t},\frac{1}{M}\}_{i=1:M}$.

Step 3: Calculate the marginal density:

$$p(y_t|y_{1:t-1};\Theta)=\sum_{s_t=1}^{S}\sum_{s_{t-1}=1}^{S}p(y_t|s_t,s_{t-1},y_{1:t-1};\Theta)\times p(s_t,s_{t-1}|y_{1:t-1};\Theta).$$

Step 4: Update regime probabilities:

$$p(s_t|y_{1:t};\Theta)=\frac{\sum_{s_{t-1}=1}^{S}p(y_t|s_t,s_{t-1},y_{1:t-1};\Theta)\times p(s_t,s_{t-1}|y_{1:t-1};\Theta)}{p(y_t|y_{1:t-1};\Theta)}.$$

end for

Algorithm 3 Kim Kalman Smooth Particle Filter (2KSPF).

Step 0: Initialization

Given Θ, set $p(s_0;\Theta)$ to the regime ergodic probabilities and draw S distributions $\{h^{(i)}_{s_0,0}, w^{(i)}_{s_0,0}=\frac{1}{M}\}_{i=1:M}$ in $\mathcal{N}(h_0;\mu_{s_0},\frac{\sigma^2_\eta}{1-\phi^2})$.

for $t=1,\ldots,T$ **do**

Step 1: Given $p(s_{t-1}|y_{1:t-1};\Theta)$, predict regime probabilities:

$$p(s_t,s_{t-1}|y_{1:t-1};\Theta)=p(s_t|s_{t-1};\Theta)\times p(s_{t-1}|y_{1:t-1};\Theta).$$

Step 2-1: Given the S distributions $\{h^{(i)}_{s_{t-1},t-1},\frac{1}{M}\}_{i=1:M}$ and the current regime s_t, compute $\{h^{(i)}_{s_t,s_{t-1},t|t-1}\}_{i=1:M}$ and $\{y^{(i)}_{s_t,s_{t-1},t|t-1}\}_{i=1:M}$ using the state-space representation:

$$h^{(i)}_{s_t,s_{t-1},t|t-1}=f(h^{(i)}_{s_{t-1},t},\xi^{(i)}_t;\Theta_{s_t}),$$
$$y^{(i)}_{s_t,s_{t-1},t|t-1}=g(h^{(i)}_{s_t,s_{t-1},t|t-1},\varepsilon^{(i)}_t;\Theta_{s_t}).$$

Step 2-2: Compute empirical moments for $x=h,y$:

$$x_{s_t,s_{t-1},t|t-1}=\frac{1}{M}\sum_{i=1}^{M}x^{(i)}_{s_t,s_{t-1},t|t-1},$$
$$P_{x_{s_t,s_{t-1},t|t-1}}=\frac{1}{M}\sum_{i=1}^{M}(x^{(i)}_{s_t,s_{t-1},t|t-1}-x_{x_t,s_{t-1},t|t-1})^2,$$
$$P_{y,h_{s_t,s_{t-1},t|t-1}}=\frac{1}{M}\sum_{i=1}^{M}(y^{(i)}_{s_t,s_{t-1},t|t-1}-y_{s_t,s_{t-1},t|t-1})(h^{(i)}_{s_t,s_{t-1},t|t-1}-h_{s_t,s_{t-1},t|t-1}).$$

Step 2–3: Deduce the Kalman gain:

$$K_{s_t,s_{t-1},t}=P_{y,h_{s_t,s_{t-1},t|t-1}}(P_{y_{s_t,s_{t-1},t|t-1}})^{-1},$$

and implement the Kalman updating step:

$$h_{s_t,s_{t-1},t|t}=h_{s_t,s_{t-1},t|t-1}+K_{s_t,s_{t-1},t}(y_t-y_{s_t,s_{t-1},t|t-1}),$$
$$P_{h_{s_t,s_{t-1},t|t}}=P_{h_{s_t,s_{t-1},t|t-1}}-K_{s_t,s_{t-1},t}P_{y_{s_t,s_{t-1},t|t-1}}(K_{s_t,s_{t-1},t})'.$$

Step 2–4: Draw current particles $\tilde{h}^{(i)}_{s_t,s_{t-1},t}$ in the proposal $\mathcal{N}(h_{s_t,s_{t-1},t|t},P_{h_{s_t,s_{t-1},t|t}})$ with weights

$$\hat{w}^{(i)}_{s_t,s_{t-1},t}\propto\frac{1}{M}\times p(y_t|\tilde{h}^{(i)}_{s_t,s_{t-1},t};\Theta_{s_t})\frac{\mathcal{N}(\tilde{h}^{(i)}_{s_t,s_{t-1},t};h_{s_t,s_{t-1},t|t-1},P_{h_{s_t,s_{t-1},t|t-1}})}{\mathcal{N}(\tilde{h}^{(i)}_{s_t,s_{t-1},t};h_{s_t,s_{t-1},t|t},P_{h_{s_t,s_{t-1},t|t}})}.$$

Step 2–5: Calculate the conditional density:

$$p(y_t|s_t,s_{t-1},y_{1:t-1};\Theta)\approx\sum_{i=1}^{M}\hat{w}^{(i)}_{s_t,s_{t-1},t}.$$

Step 2–6: $\forall s_t\in\{1,\ldots,S\}$, build $\left\{\tilde{h}^{(i)}_{s_t,s_{t-1},t},\frac{p(s_t,s_{t-1}|y_{1;t-1};\Theta)\times\hat{w}^{(i)}_{s_t,s_{t-1},t}}{\sum_{i=1}^{M}\sum_{s_{t-1}=1}^{S}p(s_t,s_{t-1}|y_{1;t-1};\Theta)\times\hat{w}^{(i)}_{s_t,s_{t-1},t}}\right\}_{i=1:M,s_{t-1}=1:S}$ andsmooth resample M particles to obtain the distribution $\{h^{(i)}_{s_t,t},\frac{1}{M}\}_{i=1:M}$.

Step 3: Calculate the marginal density:

$$p(y_t|y_{1:t-1};\Theta)=\sum_{s_t=1}^{S}\sum_{s_{t-1}=1}^{S}p(y_t|s_t,s_{t-1},y_{1:t-1};\Theta)\times p(s_t,s_{t-1}|y_{1:t-1};\Theta).$$

Step 4: Update regime probabilities:

$$p(s_t|y_{1:t};\Theta)=\frac{\sum_{s_{t-1}=1}^{S}p(y_t|s_t,s_{t-1},y_{1:t-1};\Theta)\times p(s_t,s_{t-1}|y_{1:t-1};\Theta)}{p(y_t|y_{1:t-1};\Theta)}.$$

end for

Algorithm 4 : LMDZ ((Langrock et al., 2012)).

Step 0: Initialization

Given Θ, build a fixed grid $\{h_i\}_{i=1:M}$ by creating equidistant nodes on $\left[\mu_1 - 2.5\frac{\sigma_\eta}{\sqrt{1-\phi^2}}, \mu_2 + 2.5\frac{\sigma_\eta}{\sqrt{1-\phi^2}}\right]$ (with $\mu_1 < \mu_2$).

Set $p(s_0;\Theta)$ to the regime ergodic probabilities and use $p(h_0 = h_i|s_0;\Theta) = \frac{1}{M}$ as a discrete approximation of $h_0|s_0;\Theta$.

for $t = 1,\ldots,T$ **do**

Step 1: Given $p(s_{t-1}|y_{1:t-1};\Theta)$, predict regime probabilities:

$$p(s_t, s_{t-1}|y_{1:t-1};\Theta) = p(s_t|s_{t-1};\Theta) \times p(s_{t-1}|y_{1:t-1};\Theta).$$

Step 2: Following [8], simulate the state equation for $t-1$ with the grid $\{h_i\}_{i=1:M}$ and use [61] to compute the M^2 transition probabilities $p\big(h_t = h_j|h_{t-1} = h_i, s_t, s_{t-1}, y_{1:t-1};\Theta\big)$.

Step 3: Given $p(h_{t-1} = h_i|s_{t-1}, y_{1:t-1};\Theta)$, deduce

$$p(h_t = h_j, h_{t-1} = h_i|s_t, s_{t-1}, y_{1:t-1};\Theta) = p(h_t = h_j|h_{t-1} = h_i, s_t, s_{t-1}, y_{1:t-1};\Theta) \times p(h_{t-1} = h_i|s_{t-1}, y_{1:t-1};\Theta),$$

$$p(h_t = h_j, h_{t-1} = h_i, s_t, s_{t-1}|y_{1:t-1};\Theta) = p(h_t = h_j, h_{t-1} = h_i|s_t, s_{t-1}, y_{1:t-1};\Theta) \times p(s_t, s_{t-1}|y_{1:t-1};\Theta).$$

Step 4: From the model, compute the conditional likelihood:

$$p(y_t|h_t = h_j, h_{t-1} = h_i, s_t, s_{t-1}, y_{1:t-1};\Theta).$$

Step 5: Calculate the joint, conditional and marginal densities of observation y_t:

$$p(y_t, h_t = h_j, h_{t-1} = h_i, s_t, s_{t-1}|y_{1:t-1};\Theta) = p(y_t|h_t = h_j, h_{t-1} = h_i, s_t, s_{t-1}, y_{1:t-1};\Theta) \times p\big(h_t = h_j, h_{t-1} = h_i, s_t, s_{t-1}|y_{1:t-1};\Theta\big),$$

$$p(y_t|s_t, s_{t-1}, y_{1:t-1};\Theta) = \sum_{i=1}^{M}\sum_{j=1}^{M} p(y_t, h_t = h_j, h_{t-1} = h_i, s_t, s_{t-1}|y_{1:t-1};\Theta),$$

$$p(y_t|y_{1:t-1};\Theta) = \sum_{s_t=1}^{S}\sum_{s_{t-1}=1}^{S} p(y_t|s_t, s_{t-1}, y_{1:t-1};\Theta) \times p(s_t, s_{t-1}|y_{1:t-1};\Theta).$$

Step 6: As in [48], implement the collapsing technique:

$$p(y_t, h_t = h_j, s_t|y_{1:t-1};\Theta) = \sum_{i=1}^{M}\sum_{s_{t-1}=1}^{S} p(y_t, h_t = h_j, h_{t-1} = h_i, s_t, s_{t-1}|y_{1:t-1};\Theta),$$

$$p(h_t = h_i|s_t, y_{1:t};\Theta) = \frac{p(y_t, h_t = h_j, s_t|y_{1:t-1};\Theta)}{\sum_{j=1}^{M} p(y_t, h_t = h_j, s_t|y_{1:t-1};\Theta)}.$$

Step 7: Update regime probabilities:

$$p(s_t|y_{1:t};\Theta) = \frac{\sum_{s_{t-1}=1}^{S} p(y_t|s_t, s_{t-1}, y_{1:t-1};\Theta) \times p(s_t, s_{t-1}|y_{1:t-1};\Theta)}{p(y_t|y_{1:t-1};\Theta)}.$$

end for

Algorithm 5 The online filter for the MS-SV model ((Carvalho and Lopes, 2007)).

Step 0: Initialization
Set $\delta \in [0.9, 0.99]$ and set $b^2 = 1 - (\frac{3\delta-1}{2\delta})^2$ and $a = \sqrt{1-b^2}$. Given Θ, draw $\{s_0^{(i)}\}_{i=1:M}$ in the regime ergodic probabilities distribution $p(s_0;\Theta)$; draw $\{h_0^{(i)}\}_{i=1:M}$ in the S-Gaussian mixture $\sum_{s_0=1}^{S} p(s_0;\Theta) \times \mathcal{N}(h_0; \mu_{s_0}, \frac{\sigma_\eta^2}{1-\phi^2})$. Define $\{\Theta_0^{(i)}\}_{i=1:M}$. Set all weights to $w_0^{(i)} = \frac{1}{M}$.

for $t = 1, \ldots, T$ **do**

Step 1: Given $\{h_{t-1}^{(i)}, s_{t-1}^{(i)}, \Theta_{t-1}^{(i)}, w_{t-1}^{(i)}\}_{i=1:M}$, calculate $\bar{\Theta}_{t-1} = \sum_{i=1}^{M} w_{t-1}^{(i)} \Theta_{t-1}^{(i)}$ and $V_{t-1} = \sum_{i=1}^{M} w_{t-1}^{(i)} (\Theta_{t-1}^{(i)} - \bar{\Theta}_{t-1})(\Theta_{t-1}^{(i)} - \bar{\Theta}_{t-1})'$ .Define the predictions for the state variables:

$$\bar{\Theta}_t^{(i)} = a\Theta_{t-1}^{(i)} + (1-a)\bar{\Theta}_{t-1},$$
$$\bar{s}_t^{(i)} = \arg\max_k p(s_t = k | s_{t-1}^{(i)}; \bar{\Theta}_t^{(i)}),$$
$$\bar{h}_t^{(i)} = \mathbb{E}[p(h_t | h_{t-1}^{(i)}, \bar{s}_t^{(i)}; \bar{\Theta}_t^{(i)})].$$

Step 2: Select the most promising particles $\{h_{t-1}^{(i)}, s_{t-1}^{(i)}, \Theta_{t-1}^{(i)}\}_{i=1:M}$ by sampling the index k^i from $\tilde{\tau}_{t-1}^{(i)} = \frac{w_{t-1}^{(i)} p(y_t | \bar{h}_t^{(i)}, \bar{s}_t^{(i)}; \bar{\Theta}_t^{(i)})}{\sum_{i=1}^{M} w_{t-1}^{(i)} p(y_t | \bar{h}_t^{(i)}, \bar{s}_t^{(i)}; \bar{\Theta}_t^{(i)})}$.

Step 3: Draw current particles:

$$\tilde{\Theta}_t^{(i)} \text{ from } \mathcal{N}(\bar{\Theta}_t^{(k^i)}, b^2 V_{t-1}),$$
$$\tilde{s}_t^{(i)} \text{ from } p(s_t | s_{t-1}^{(k^i)}; \tilde{\Theta}_t^{(i)}),$$
$$\tilde{h}_t^{(i)} \text{ from } p(h_t | h_{t-1}^{(k^i)}, \tilde{s}_t^{(i)}; \tilde{\Theta}_t^{(i)}).$$

Step 4: Calculate the unormalized weights:

$$\hat{w}_t^{(i)} \propto p(y_t | \tilde{h}_t^{(i)}, \tilde{s}_t^{(i)}; \tilde{\Theta}_t^{(i)}) \frac{w_{t-1}^{(k^i)}}{\tilde{\tau}_{t-1}^{(k^i)}}.$$

Step 5: Resample $\{\tilde{h}_t^{(i)}, \tilde{s}_t^{(i)}, \tilde{\Theta}_t^{(i)}, \tilde{w}_t^{(i)} = \frac{\hat{w}_t^{(i)}}{\sum_{i=1}^{M} \hat{w}_t^{(i)}}\}_{i=1:M}$ to obtain $\{h_t^{(i)}, s_t^{(i)}, \Theta_t^{(i)}, w_t^{(i)}\}_{i=1:M}$.

end for

Supplementary material

Supplementary material associated with this article can be found, in the online version, at 10.1016/j.ecosta.2018.05.004 .

References

Abanto-Valle, C., Migon, H., Lachos, V., 2011. Stochastic volatility in mean models with scale mixtures of normal distributions and correlated errors: a Bayesian approach. J. Stat. Plann. Inf. 141, 1875–1887.
Albertini, J., Lan, H., 2016. The Importance of Time-varying Parameters in New Keynesian Model with Zero Lower Bound.
Alstadheim, R., Bjørnland, H., Maih, J., 2013. Do Central Banks Respond to Exchange Rate Movements? A Markov-Switching Structural Investigation. Working Paper. Norges Bank Research.
Arulampalam, S., Maskell, S., Gordon, N., Clapp, T., 2002. A tutorial on particle filters for on-line non-linear / non gaussian Bayesian tracking. IEEE Trans. Signal Process 50, 241–254.
Bianchi, F., 2013. Regime switches, agents' beliefs, and post-world war II U.S. macroeconomic dynamics. Rev. Econ. Stud. 80 (2), 463–490.
Bilio, M., Casarin, R., 2010. Identifying business cycle turning points with sequential monte carlo methods: an online and real-time application to the Euro area. J. Forecast. 29, 145–167.
Bilio, M., Casarin, R., Sartore, D., 2010. Bayesian Inference on Dynamic Models with Latent Factors. EUROSTAT.
Binning, A., Maih, J., 2015. Sigma Point Filters For Dynamic Nonlinear Regime Switching Models. Working Paper. CAMP Working Paper Series.
Binning, A., Maih, J., 2016. Implementing the Zero Lower Bound in an Estimated Regime-Switching DSGE Model. Working Paper. Norges Bank Research.
Bollerslev, T., 1986. Generalized autoregressive conditional heteroskedasticity. J. Econ. 31, 307–327.
Bollerslev, T., Kretschmer, U., Pigorsch, C., Tauchen, G., 2009. A discrete-time model for daily SP500 returns and realized variations: jumps and leverage effects. J. Econ. 150, 151–166.
Cai, J., 1994. A Markov model of switching-regime ARCH. J. Bus. Econ. Stat. 12, 309–316.
Cappe, O., Godsill, S., Moulines, E., 2007. An overview of existing methods and recent advances in sequential monte carlo. Proc. IEEE 95 (5), 899–924.
Carvalho, C., Lopes, H., 2007. Simulation-based sequential analysis of Markov-switching stochastic volatility models. Comput. Stat. Data Anal. 51, 4526–4542.
Chen, Z., 2003. Bayesian filtering: from Kalman filters to particle filters, and Beyond. Statistics 182, 1–69.
Chen, Z., Liu, J., 2000. Mixture Kalman filters. J. R. Stat. Soc. (B) 62, 493–508.
Chib, S., Nardari, F., Shephard, N., 2002. Markov chains Monte Carlo methods for stochastic volatility models. J. Econ. 108, 281–316.
Chopin, N., Jacob, P., Papaspiliopoulos, O., 2013. SMC2: an efficient algorithm for sequential analysis of state-space models. J. R. Stat. Soc. Ser. B 75 (3), 397–426.
Creal, D., 2009. A survey of sequential monte carlo methods for economics and finance. Econ. Rev. 31, 245–296.
Crisan, D., Miguez, J., 2018. Nested particle filters for online parameter estimation in discrete-time state-space Markov models. Bernoulli 24 (4A), 3039–3086.
Diebold, F., Mariano, R., 1995. Comparing predictive accuracy. J. Bus. Econ. Stat. 13 (3), 134–144.
Doucet, A., de Freitas, N., Gordon, N. (Eds.), 2001. Sequential Monte Carlo Methods in Practice. New-York Springer.
Doucet, A., Johansen, A., 2009. A tutorial on particle filtering and smoothing: fifteen years later. Handbook of Nonlinear Filtering.
Dueker, M., Fischer, A., Dittmar, R., 2002. Stochastic Capital Depreciation and the Comovement of Hours and Productivity. Working Paper. Federal Reserve Bank of St. Louis.
Engle, R., 1982. Autoregressive conditional heteroskedasticity with estimates of the variance of UK inflation. Econometrica 50, 987–1008.
Fearnhead, P., Clifford, P., 2003. On-line inference for hidden Markov models via particle filters. J. R. Stat. Soc. (B) 65 (4), 887–899.
Foerster, A., Rubio-Ramírez, J., Waggoner, D., Zha, T., 2016. Perturbation methods for Markov-switching dynamic stochastic general equilibrium models. Quant. Econ. 7 (2), 637–669.
Gallant, A., Hsieh, D., Tauchen, G., 1997. Estimation of stochastic volatility models with diagnostics. J. Econ. 81, 159–192.
Gallant, A., Tauchen, G., 1996. Which moments to match. Econ. Theory 12, 657–681.
Gerber, M., Chopin, N., 2015. Sequential Quasi Monte Carlo. J. R. Stat. Soc. Ser. B (Statistical Methodology) 77 (3), 509–579.
Gordon, N., Salmond, D., Smith, A., 1993. Novel approach to nonlinear and non-Gaussian Bayesian state estimation. IEEE Proc.-F 140, 107–113.
Gourieroux, C., Monfort, A., Renault, E., 1993. Indirect inference. J. Appl. Econ. 8, 85–118.
Hamilton, J., Susmel, R., 1994. Autoregressive conditional heteroskedasticity and changes in Regime. J. Econ. 64, 307–333.
Harvey, A., Ruiz, E., Shephard, N., 1994. Multivariate stochastic variance models. Rev. Econ. Stud. 61 (2), 247–264.
Heston, S., 1993. A closed-form solution for options with stochastic volatility with applications to bond and currency options. Rev. Financial Stud. 6, 327–343.
Hull, J., White, A., 1987. The pricing of options on assets with stochastic volatilities. J. Finance 42, 281–300.
Hwang, S., Satchell, S., Valls Pereira, P., 2004. Stochastic volatility models with Markov regime switching state equations. J. Bus. Econ. Stat. 16.
Jacquier, E., Polson, N., Rossi, P., 1994. Bayesian analysis of stochastic volatility models. J. Bus. Econ. Stat. 12 (4), 69–87.
Johannes, M., Polson, N., 2002. MCMC methods for financial econometrics. In: Handbook of Financial Econometrics.
Kantas, N., Doucet, A., Singh, S., Maciejowski, J., Chopin, N., 2015. On particle methods for parameter estimation in state-space models. Stat. Sci. 30 (3), 328–351.
Kim, C., 1994. Dynamic linear models with Markov-switching. J. Econ. 60, 1–22.
Kim, C., Nelson, C., 1999. State-Space Models with Regime Switching: Classical and Gibbs Sampling Approaches with Applications. MIT Press.
Kim, J., 2016. Efficient Bayesian Inference in Non-linear Switching State Space Models Using Particle Gibbs Sampling Approaches. Technical Report. Working paper, University of Oklahoma.
Kotecha, J., Djuric, P., 2003. Gaussian particle filtering. IEEE Trans. Signal Process. 51 (10), 2592–2601.
Lamoureux, C., Lastrapes, W., 1990. Persistence in variance, structural change, and the GARCH Model. J. Bus. Econ. Stat. 8, 225–234.
Langrock, R., MacDonald, I., Zucchini, W., 2012. Some nonstandard stochastic volatility models and their estimation using structured hidden Markov models. J. Emp. Finance 19, 147–161.
Lee, A., 2008. Towards Smooth Particle Filters for Likelihood Estimation with Multivariate Latent Variables. University of British Columbia.
Liesenfeld, R., Richard, J., 2003. Univariate and multivariate stochastic volatility models: estimation and diagnostics. J. Emp. Finance 10 (4), 505–531.
Liesenfeld, R., Richard, J., 2006. Classical and Bayesian analysis of univariate and multivariate stochastic volatility models. Econ. Rev. 25 (2–3), 335–360.
Liu, J., West, M., 2001. Combined parameter and state estimation in simulation-based filtering. Sequential Monte Carlo Methods in Practice.
Liu, P., Mumtaz, H., 2011. Evolving macroeconomic dynamics in a small open economy: an estimated markov switching DSGE model for the UK. J. Money Credit Bank. 43 (7), 1443–1474.
Liu, Z., Waggoner, D., Zha, T., 2011. Sources of macroeconomic fluctuations: a regime-switching DSGE approach. Quant. Econ. 2 (2), 251–301.
Malik, S., Pitt, M., 2011. Particle filters for continuous likelihood evaluation and maximisation. J. Econ. 165 (2), 190–209.
Murray, L., Jones, E., Parslow, J., 2013. On disturbance state-space models and the particle marginal metropolis-hastings sampler. J. Uncertainty Quant. 1 (1), 494–521.
Nakajima, J., Omori, Y., 2009. Leverage, heavy-tails and correlated jumps in stochastic volatility models. Comput. Stat. Data Anal. 53, 2335–2353.
Omori, Y., Chib, S., Shephard, N., Nakajima, J., 2007. Stochastic volatility with leverage: fast likelihood inference. J. Econ. 140, 425–449.
Pitt, M., 2002. Smooth Particle Filters for Likelihood Evaluation and Maximisation. Technical Report. Warwick Economic Research Papers, University of Warwick.

Pitt, M., Shephard, N., 1999. Filtering via simulation: auxiliary particle filters. J. Am. Stat. Assoc. 94 (446), 590–599.
Raggi, D., Bordignon, S., 2006. Comparing stochastic volatility models through monte carlo simulations. Comput. Stat. Data Anal. 50, 1678–1699.
Sandmann, G., Koopman, S., 1998. Estimation of volatility stochastic models via Monte Carlo maximum likelihood. J. Econ. 87, 271–301.
Smith, A., 1993. Estimating non linear time series model using simulated vector autoregressions. J. Appl. Econ. 8, 63–84.
Smith, R., 2002. Markov-switching and stochastic volatility diffusion models of short-term interest rates. J. Bus. Econ. Stat. 20 (2), 183–197.
So, M., Lam, K., Li, W., 1998. A stochastic volatility model with Markov-switching. J. Bus. Econ. Stat. 16 (2), 244–253.
Tauchen, G., 1986. Finite state Markov-chain approximations to univariate and vector autoregressions. Econ. Lett. 20, 177–181.
Taylor, S., 1986. Modelling Financial Time Series. Wiley, New York.
Wiggins, J., 1987. Option values under stochastic volatility: theory and empirical estimates. J. Financial Econ. 19, 351–372.
Winschel, V., Kratzig, M., 2010. Solving, estimating and selecting nonlinear dynamic models without the curse of dimensionality. Econometrica 78 (2), 803–821.